



Lift Assembly



AIROBOTICS

1 Introduction

2 Assembly and Sub-Assemblies Procedures

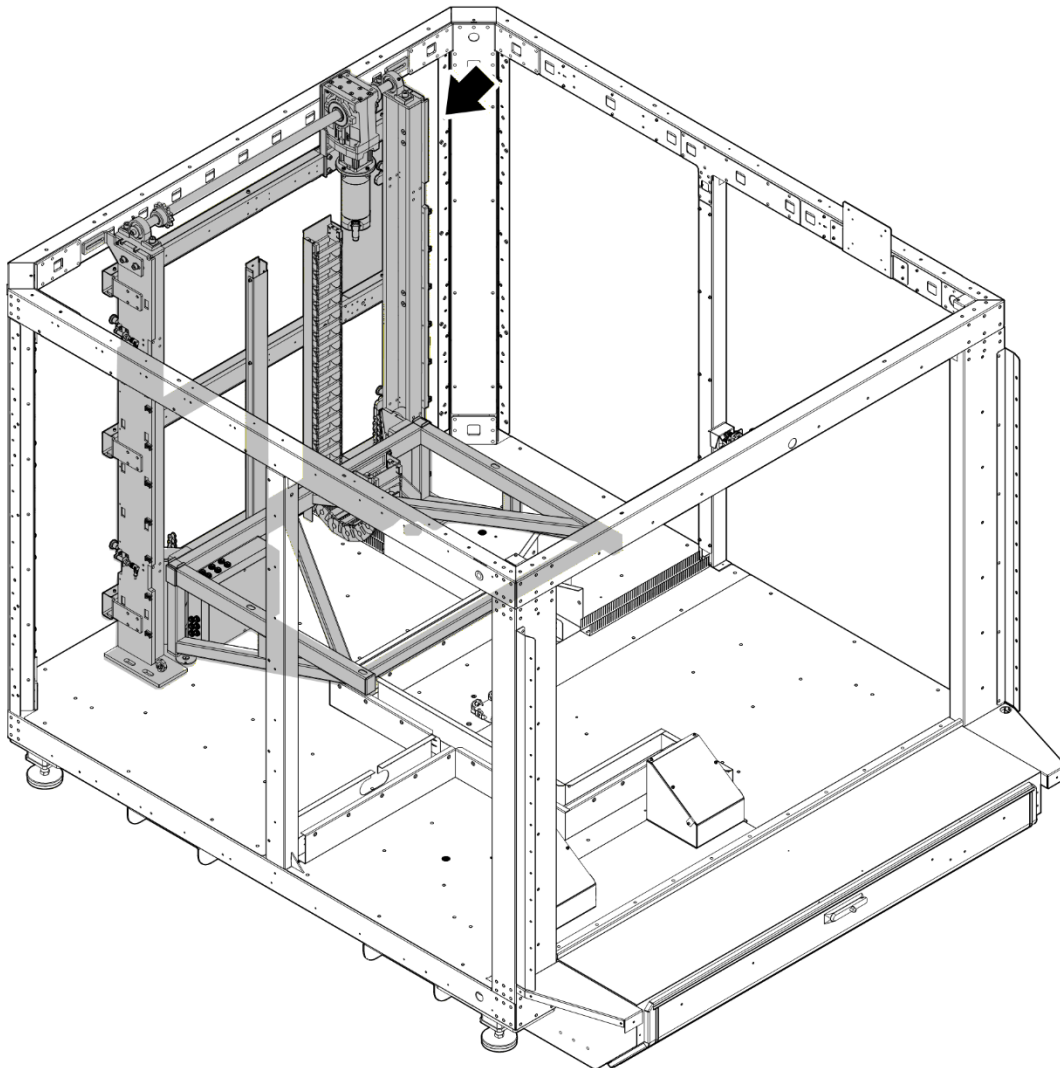
2.1 Scope

This section outlines the step-by-step procedures required for assembling the Lift assembly.

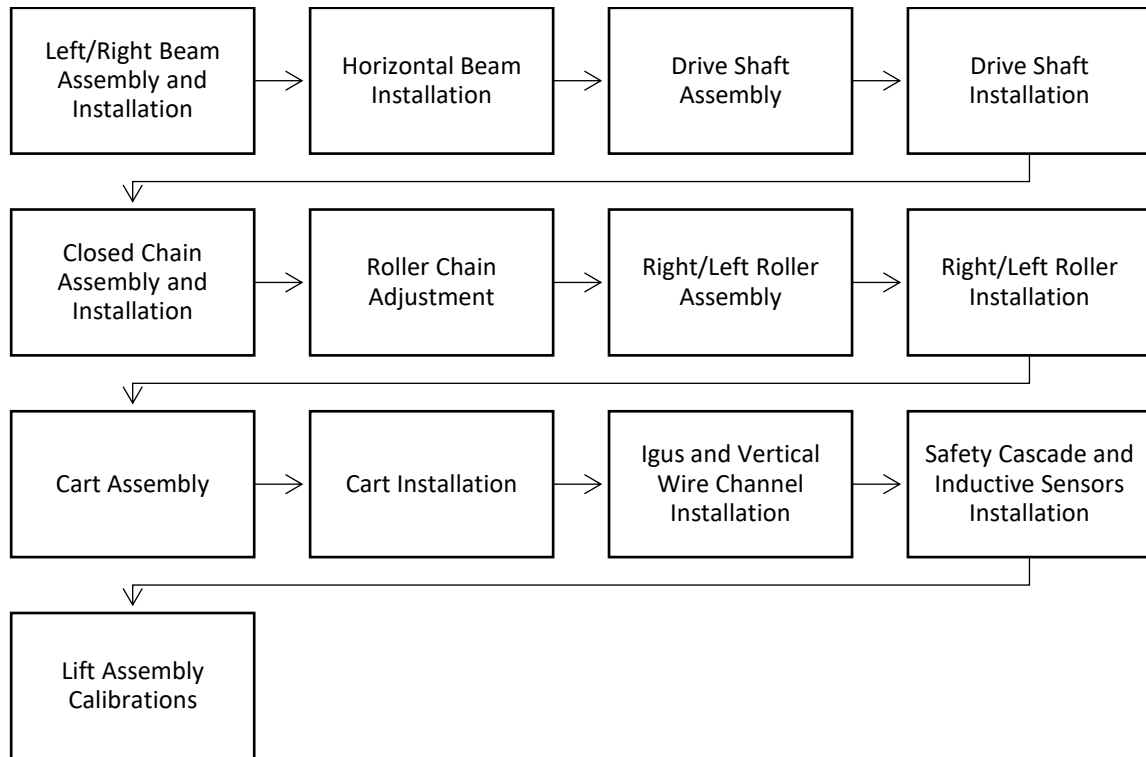
Note

- Prior to commencing assembly of each of the units described in this guide, make sure the relevant ATP documents are properly signed.
- Spread Loctite 243 or similar glue on all the screws unless stated otherwise.
- Tightening of screws will be done according to the Screw Torque Table included in this document. (See Appendix TBD).

1003-052v01



2.2 Assembly Process



2.3 Drawing List

#	Drawing P/N	Drawing Name
1	AS230145	LIFT ASSEMBLY
2	AS230135	RIGHT BEAM WELDING
3	AS230137	LEFT BEAM WELDING
4	SL230172	Top / Middle / Bottom Horizontal beam
5	MH230046	Roller Chain
6	SL230174	Left Roller Base
7	SL230184	Right Roller Base
8	AS230149	Lift Electric Box
9	AS230140	Cart Assembly
10	SL230192	Vertical Wire Channel
11	SL230191	Igus Bracket
12	IG230001	Cable Chain (Igus)



2.4 Assembly Procedures

2.4.1 Left/Right Beam Installation

2.4.1.1 Left Beam Installation

2.4.1.2 Left Beam Installation Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-1	AS230136	Left Beam	1	Component
2	Figure 2-1	FS230272	DIN912 M10x30 Screw	2	Fastener
3	Figure 2-1	FS230138	DIN9021 M10 Washer	2	Fastener
4	Figure 2-1	SL230171	Lift Bracket Left	1	Component
5	Figure 2-1	FS230374	DIN912 M10x20 Screw	2	Fastener
6	Figure 2-1	FS230146	DIN125 M10 Washer	2	Fastener
7	Figure 2-1	FS230586	ISO7380 M5X8 Screw	8	Fastener
8	Figure 2-1	LS230066	Cable Tie Holder	8	Fastener

2.4.1.3 Left Beam Installation Procedure

(Refer to Figure 2-1 below).

1. Install the electronic cabinet according to section **TBD**.
2. Assemble Left Beam Welding, refer to assembly drawing **AS230137**.
3. Position the Left Beam (1) on the UPS and FAN Electrical Assembly in its most forward position (View A) as defined by the base installation ellipse bores.

Note

Do not apply Loctite on the M10 Allen-head screws (2) which secure the Left Beam (1) to the UPS and FAN Electrical Assembly

4. Using an 8 mm Allen bit, install two M10 Allen-head screws (2) and two M10 flat washers (3) which secure the Left Beam (1) to the UPS and FAN Electrical Assembly.
5. Position the Lift Bracket (4) in its place on the Left Beam (1).

Note

Do not apply Loctite on the M10 Allen-head screws (6) which secure the Lift Bracket (4) to Left Beam (1).

6. Using an 8 mm Allen bit, install two M10 Allen-head screws (6) and two M10 washers (5), to secure the Lift Bracket (4) to the Left Beam (1).

7. Using an 8 mm Allen bit, install two Allen-head M10 screws (7) and two M10 washers (6), to secure the Lift Bracket (5) to the Left Beam (1).
8. Position eight Tie Mounts (8) on the installation bores of the Beam (1).
9. Using a 4 mm Allen-head bit, install eight M5 Allen-head screws (7) which secure eight Tie Mounts (8) to the Left Beam (1).

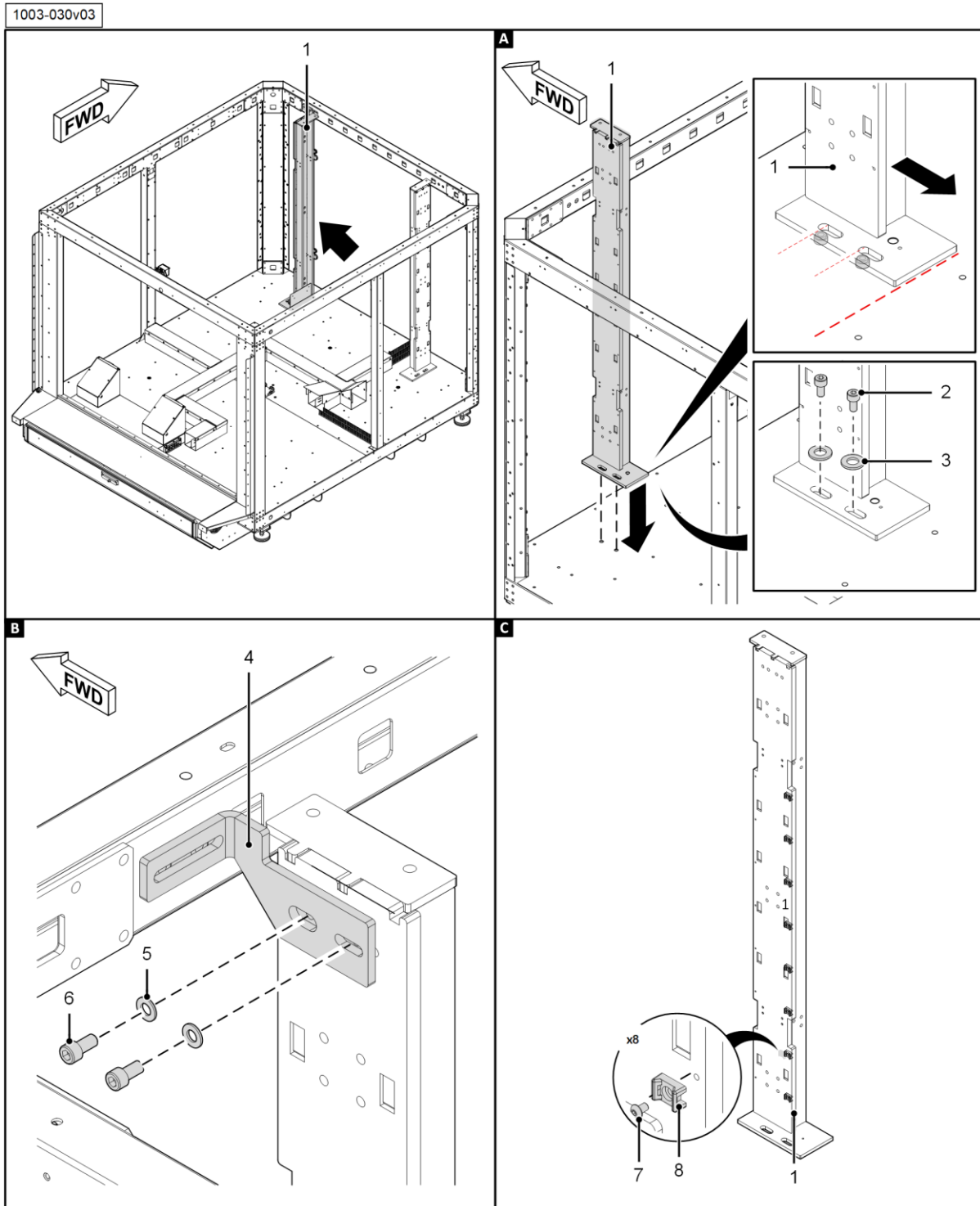


Figure 2-1: Left/Right Beam Installation

2.4.1.4 Right Beam Installation

2.4.1.5 Right Beam Installation Parts List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-3	AS230134	Right Beam	1	Component
2	Figure 2-3	LS230066	Cable Tie Holder	8	Fastener
3	Figure 2-3	FS230586	ISO7380 M5X8 Screw	8	Fastener
4	Figure 2-3	FS230272	DIN912 M10x30 Screw	2	Fastener
5	Figure 2-3	FS230138	DIN9021 M10 Washer	2	Fastener
6	Figure 2-3	SL230168	Lift Bracket Right	1	Component
7	Figure 2-1	Part of CH230024	Chassis beam	1	Component
8	Figure 2-3	FS230146	DIN125 M10 Washer	4	Fastener
9	Figure 2-3	FS230374	DIN912 M10x20 Screw	4	Fastener

2.4.1.6 Right Beam Installation Procedure

(Refer to Figure 2-2 below).

1. Assemble Right Beam Welding, refer to assembly drawing **AS230135**.
2. Position the Right Beam (1) on the UPS and FAN Electrical Assembly in its most forward position (View D) as defined by the base installation ellipse bores.

Note

Do not apply Loctite on the M10 Allen-head screws which secure the Right Beam (1) to the UPS and FAN Electrical Assembly

3. Using an 8 mm Allen bit, install two M10 Allen-head screws (4) and two M10 flat washers (5) which secure the Right Beam (1) to the UPS and FAN Electrical Assembly.
4. Position the Lift Bracket (6) in its place on the Right Beam (1).

Note

Do not apply Loctite on the M10 Allen-head screws (9) which secure the Lift Bracket (6) to the Right Beam (1)

5. Using an 8 mm Allen bit, install two M10 Allen-head screws (9) and two M10 washers (8), to secure the Lift Bracket (6) to the Right Beam (1).
6. Position eight Tie Mounts (2) on the installation bores of the Right Beam (1).
7. Using a 4 mm Allen-head bit, install eight M5 Allen-head screws (3) which secure eight Tie Mounts to the Right Beam (1).

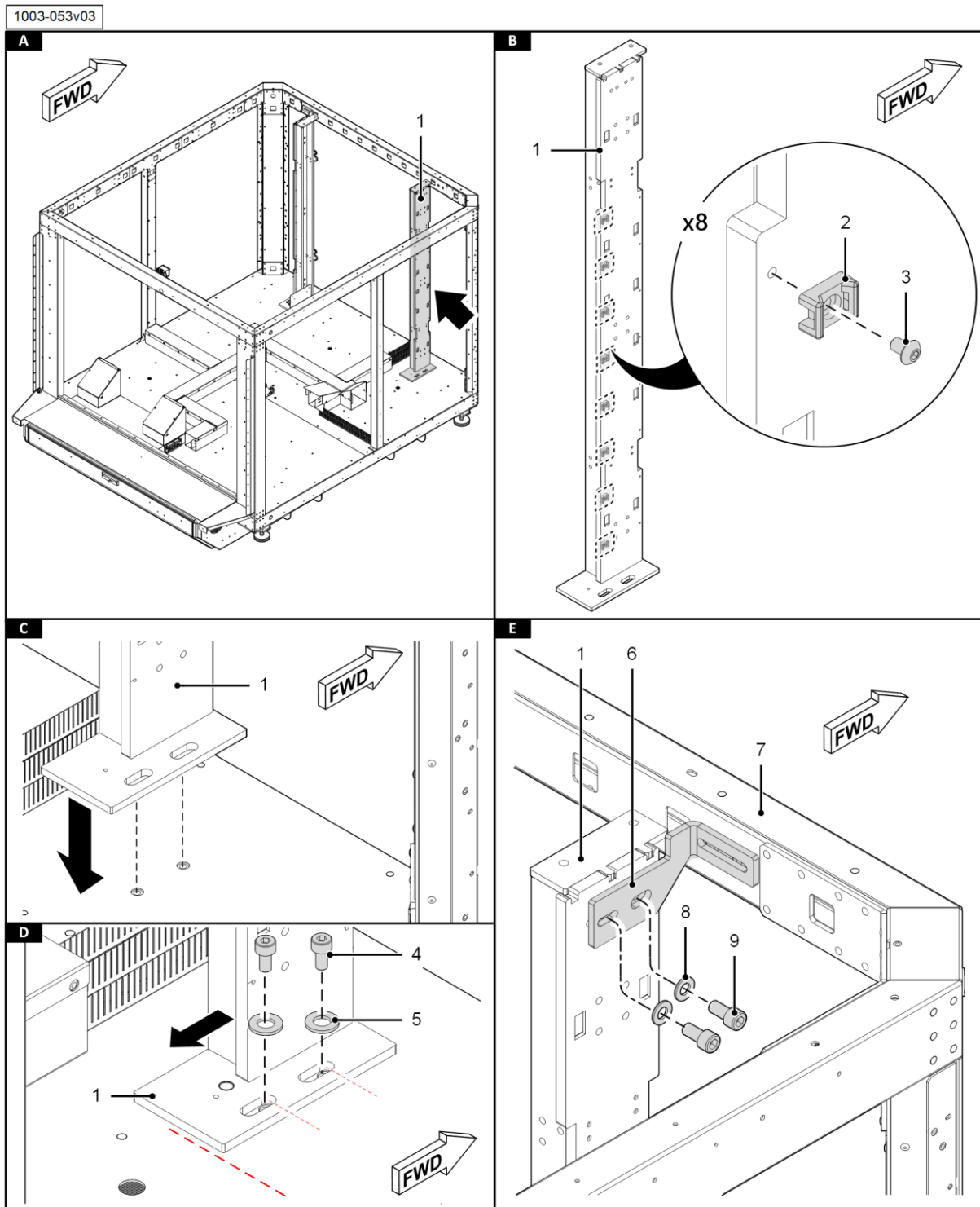


Figure 2-3: Left/Right Beam Installation

2.4.1.7 Left Index Plungers Installation

2.4.1.8 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-4	Part of MH230039	Nut	--	N/A
2	Figure 2-4	Part of MH230039	Plunger Body	--	N/A
3	Figure 2-4	Part of MH230039	Plunger Knob	--	N/A
4	Figure 2-4	MH230039	Index Plunger	2	Component
5	Figure 2-4	SL230169	Stopper	2	Component
6	Figure 2-4	AS230136	Left Beam	--	N/A
7	Figure 2-4	N/A	Stopper with Plunger	--	N/A
8	Figure 2-4	CC230079	Spacer	2	Fastener
9	Figure 2-4	FS230273	DIN127B M10 spring washer	2	Fastener
10	Figure 2-4	FS230283	DIN912 M10x30 Screw	2	Fastener
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-5	AS230136	Left Beam	--	N/A
2	Figure 2-5	SL230170	Sensor bracket	2	Component
3	Figure 2-5	FS230375	DIN7991 M4x8 screw	4	Fastener
4	Figure 2-5	EC230057	Inductive sensor 4mm	2	Component
5	Figure 2-5	Part of EC230057	Sensor Nut - Part of Inductive sensor 4mm	--	N/A
6	Figure 2-5	Part of EC230057	Locking Nut - Part of Inductive sensor 4mm	--	N/A
7	Figure 2-5	SL230169	Stopper	--	Component
8	Figure 2-5	N/A			
9	Figure 2-5	MH230040	Lifting eyebolt M6	2	Component



2.4.1.9 Installation Procedure

(Refer to Figure 2-4 below).

1. Perform Plunger Assembly as follows:
 - a. Install Plunger Nut (2) on the Plunger (3), tighten manually till end of move.
 - b. Install Securing Nut (1) on the Plunger (3), tighten manually till end of move.

Note

The Plunger Assy (4) threading must be flush on the other side of the Stopper (5).

- c. Install the plunger assy. (4) into the stopper (5) by threading it into the designated bore.
 - d. Secure the Plunger assy. (4) in the Stopper (5) by threading the Plunger Nut (1) towards the Stopper (5), tighten the Plunger Nut (1) to the Stopper (5) by hand force.
2. Repeat steps 1.a to 1.d for the second Plunger.
3. Install upper plunger as follows:
 - a. Position the Stopper with Plunger (7) in its place on the Beam (6).
 - b. Insert a Spacer (8) into installation bore.
 - c. Using an 8 mm Allen-head bit, install M10 Allen-head screw (10) and spring washer (9) which secure Stopper with Plunger (7) to the Beam (6).
4. Repeat steps 3.a to 3.c for lower plunger installation.

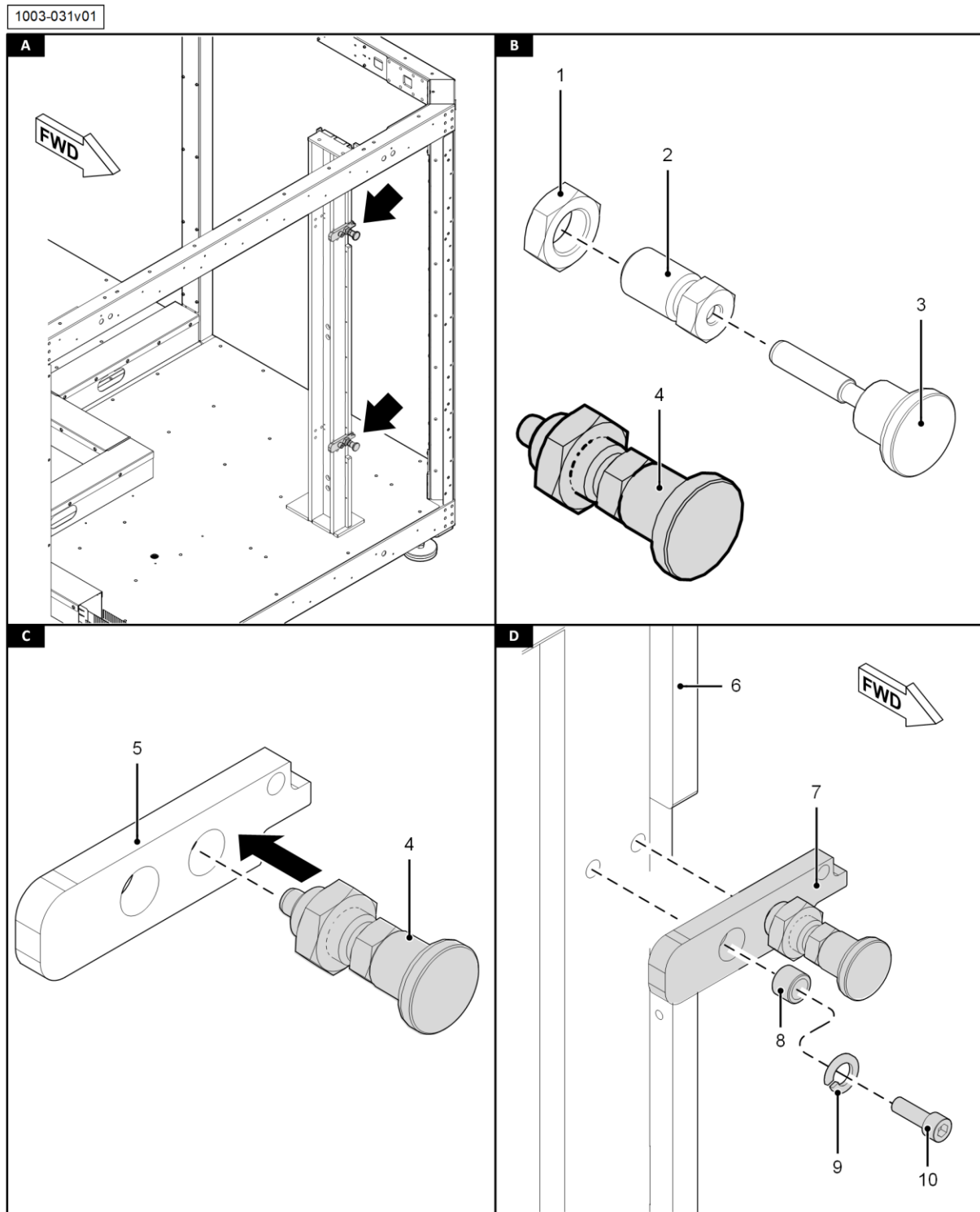


Figure 2-4: Left Index Plungers Assembly and Installation

(Refer to Figure 2-5 below).

5. Install Upper Plunger Sensor as follows:
 - a. Position upper Sensor Bracket (2) in its place on the Left Beam (1).
 - b. Using a 3 mm Allen-head bit, install two M4 Allen-head screw (3) which attach Sensor Bracket (2) to the Left Beam (1).
 - c. Thread the sensor nut (5) down until it reaches the sensor base.
 - d. Insert the Inductive Sensor (4) into Sensor Bracket (2) installation bore.
 - e. Thread the sensor locking nut (6) down until it reaches the sensor base.
 - f. Measure the distance between the Inductive Sensor tip (4) and the Stopper (6).
 - g. Adjust the distance (8) between Inductive Sensor tip (4) and the Stopper (7) to $1.25 \text{ mm} \pm 0.25 \text{ mm}$ by adjusting sensor nut (5) and locking nut (6).
 - h. Using a 17 mm open-wrench, tight the plunger sensor locking nut (6) until it reaches the Sensor Bracket (2).
6. Repeat steps 5.a to h for Lower Sensor Bracket.
7. Install Lifting Eyebolt (9) on the Left Beam (1) by threading it into its bore.

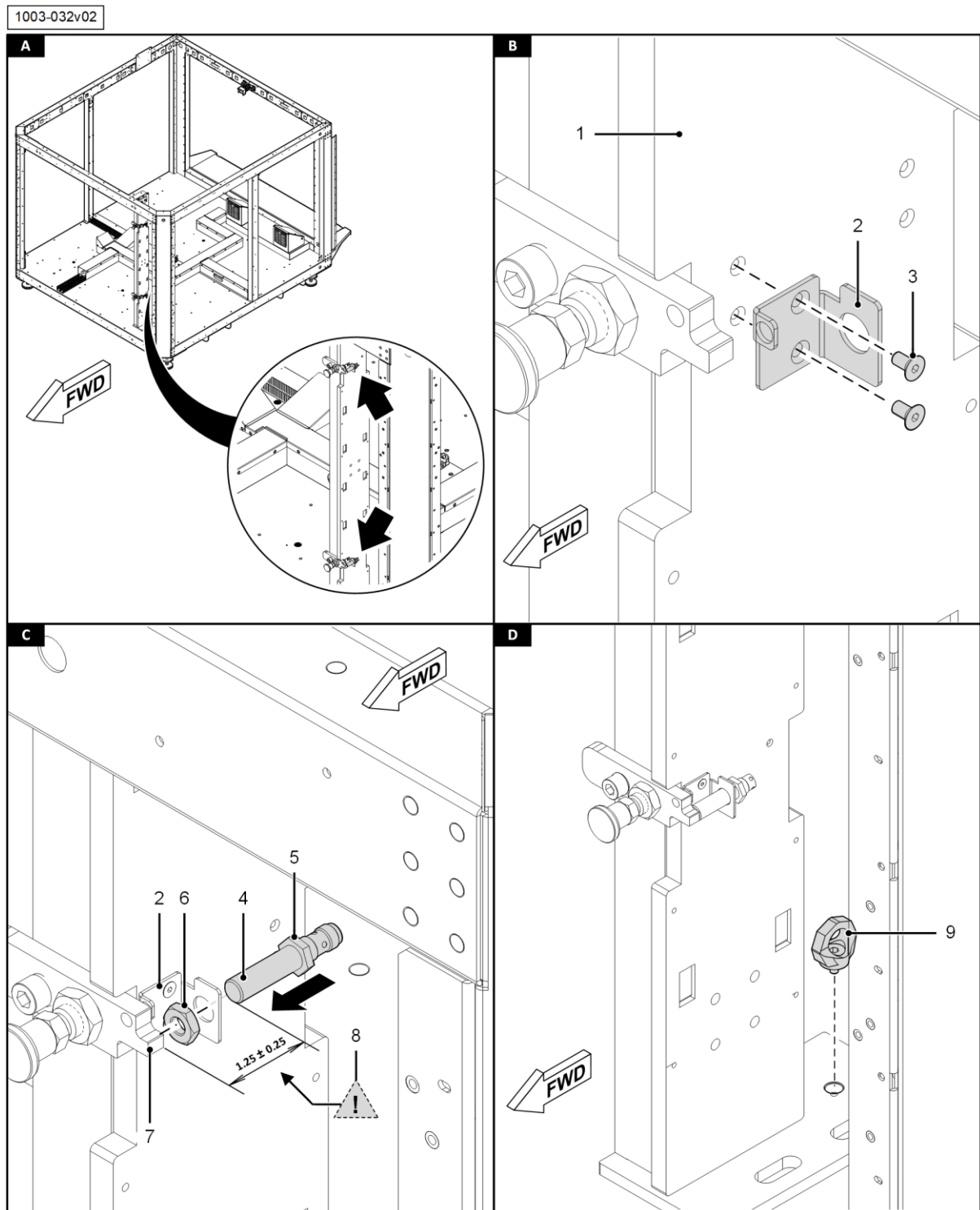


Figure 2-5: Right Sensor Installation

2.4.1.10 Right Index Plungers Installation

2.4.1.11 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-4	Part of MH230039	Nut	--	N/A
2	Figure 2-4	Part of MH230039	Plunger Body	--	N/A
3	Figure 2-4	Part of MH230039	Plunger Knob	--	N/A
4	Figure 2-4	MH230039	Index Plunger	2	Component
5	Figure 2-4	SL230169	Stopper	2	Component
6	Figure 2-4	AS230134	Right Beam	--	N/A
7	Figure 2-4	N/A	Stopper with Plunger	--	N/A
8	Figure 2-4	CC230079	Spacer	2	Fastener
9	Figure 2-4	FS230273	DIN127B M10 spring washer	2	Fastener
10	Figure 2-4	FS230283	DIN912 M10x30 Screw	2	Fastener
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-5	AS230134	Right Beam	--	N/A
2	Figure 2-5	SL230170	Sensor bracket	2	Component
3	Figure 2-5	FS230375	DIN7991 M4x8 screw	4	Fastener
4	Figure 2-5	EC230057	Inductive sensor 4mm	2	Component
5	Figure 2-5	Part of EC230057	Sensor Nut - Part of Inductive sensor 4mm	--	N/A
6	Figure 2-5	Part of EC230057	Locking Nut - Part of Inductive sensor 4mm	--	N/A
7	Figure 2-5	SL230169	Stopper	--	Component
8	Figure 2-5	N/A			
9	Figure 2-5	MH230040	Lifting eyebolt M6	2	Component

2.4.1.12 Installation Procedure

(Refer to Figure 2-6 below).

1. Perform Plunger Assembly as follows:
 - a. Install Plunger Nut (2) on the Plunger (3), tighten manually till end of move.
 - b. Install Securing Nut (1) on the Plunger (3), tighten manually till end of move.

Note

The Plunger Assy (4) threading must be flush on the other side of the Stopper (5).

- c. Install the plunger assy. (4) into the stopper (5) by threading it into the designated bore.
 - d. Secure the Plunger assy. (4) in the Stopper (5) by threading the Plunger Nut (1) towards the Stopper (5), tighten the Plunger Nut (1) to the Stopper (5) by hand force.
2. Repeat steps 1.a to 1.d for the second Plunger.
3. Install upper plunger as follows:
 - a. Position the Stopper with Plunger (7) in its place on the Right Beam (6).
 - b. Insert a Spacer (8) into installation bore.
 - c. Using an 8 mm Allen-head bit, install M10 Allen-head screw (10) and spring washer (9) which secure Stopper with Plunger (7) to the Right Beam (6).
4. Repeat steps 3.a to 3.c for lower plunger installation.



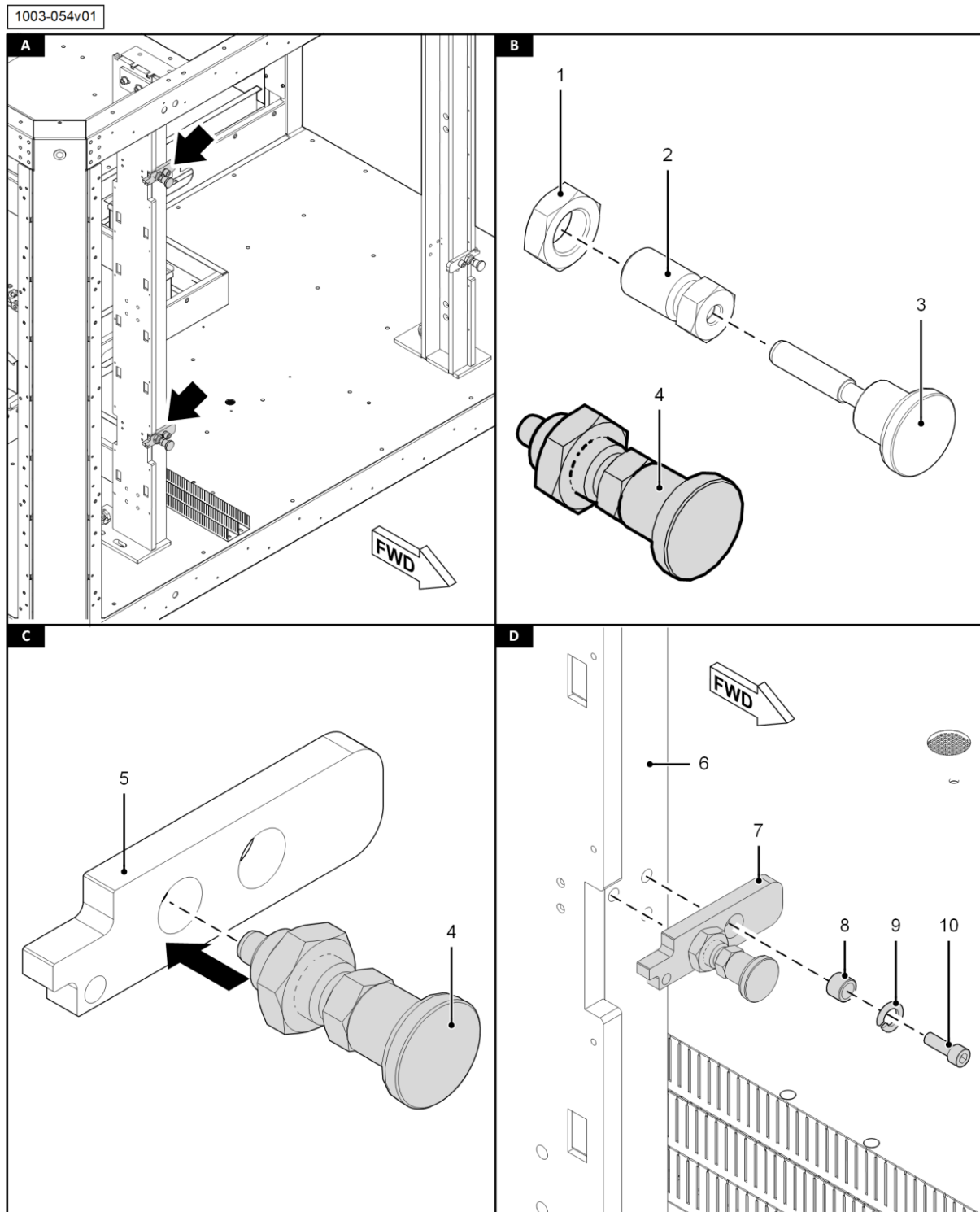


Figure 2-6: Right Index Plunger Assembly and Installation

(Refer to Figure 2-7 below).

5. Install Upper Plunger Sensor as follows:
 - a. Position upper Sensor Bracket (2) in its place on the Right Beam (1).
 - b. Using a 3 mm Allen-head bit, install two M4 Allen-head screw (3) which attach Sensor Bracket (2) to the Right Beam (1).
 - c. Thread the sensor nut (4) on the Inductive Sensor (5) until it reaches the sensor base.
 - d. Insert the Inductive Sensor (5) into Sensor Bracket (2) installation bore.
 - e. Thread the sensor locking nut (6) down until it reaches the Sensor Bracket (2).
 - f. Measure the distance (7) between the Inductive Sensor tip (5) and the Stopper (2).
 - g. Adjust the distance (7) between Inductive Sensor tip (5) and the Stopper (8) to $1.25 \text{ mm} \pm 0.25 \text{ mm}$ by adjusting sensor nut (4) and locking nut (7).
 - h. Using a 17 mm open-wrench, tight the plunger sensor locking nut (7) until it reaches the Sensor Bracket (2).
6. Repeat steps 5.a to h for Lower Sensor Bracket.
7. Install Lifting Eyebolt (9) on the Right Beam (1) by threading it into its bore.

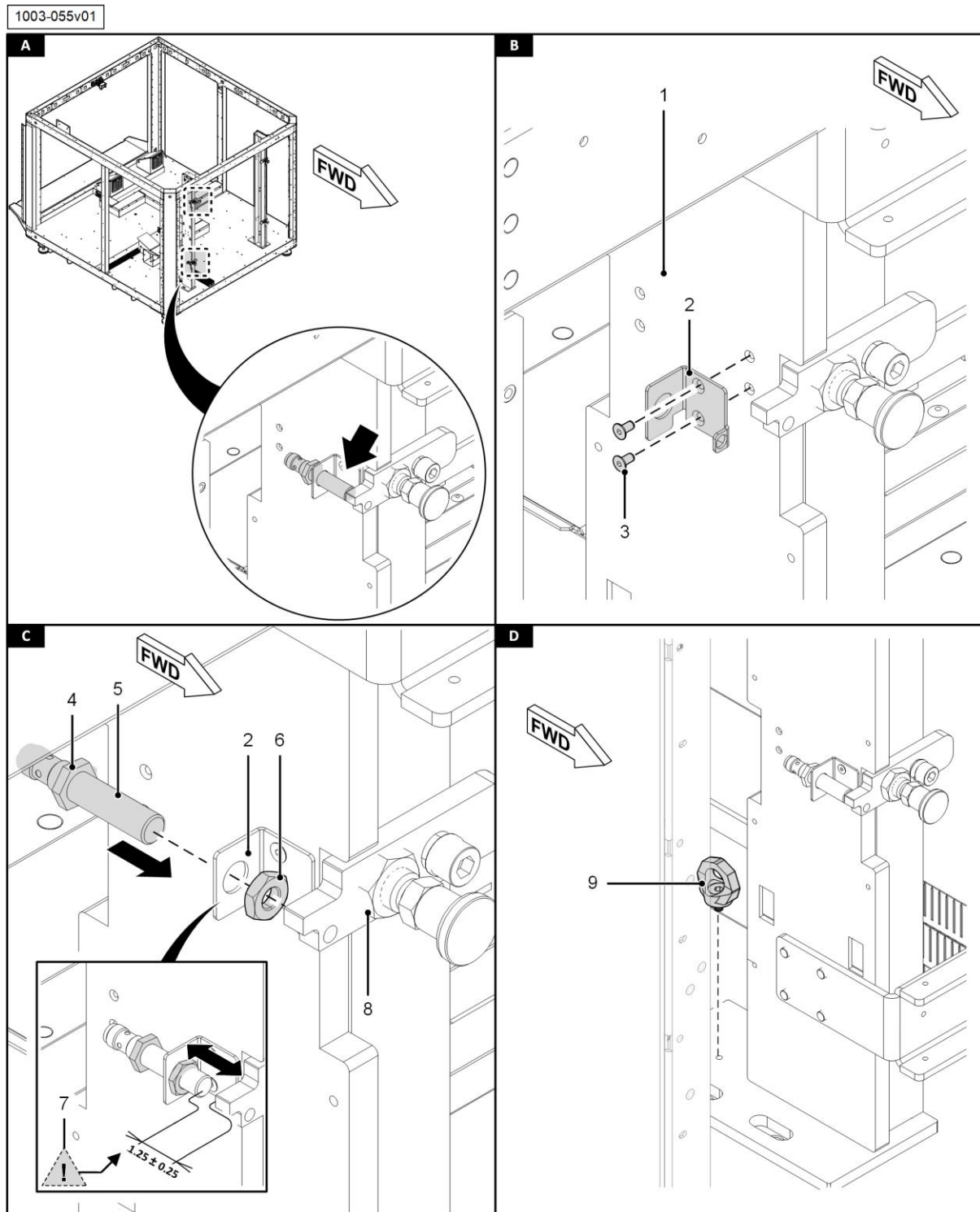


Figure 2-7: Right Sensor Installation

2.4.2 Horizontal Beams

2.4.2.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-8	SL230172	Top Horizontal beam	1	Component
2	Figure 2-8	SL230172	Middle Horizontal beam	1	Component
3	Figure 2-8	SL230172	Bottom Horizontal beam	1	Component
4	Figure 2-8	AS230134	Right Beam	--	N/A
5	Figure 2-8	AS230136	Left Beam	--	N/A
6	Figure 2-8	FS230376	DIN7991 M8x25 screw	24	Fastener

2.4.2.2 Installation Procedure

(Refer to Figure 2-8 below).

1. Position Bottom Horizontal Beam (3) in its place on the Left (5) and Right (4) Beams.
2. Using a 6 mm Allen-head bit, install eight (four on each side) M6 Allen-head screws (6) which secure Bottom Horizontal Beam (3) to the Left (5) and Right (4) Beams.
3. Repeat steps 1 to 2 for Middle Horizontal Beam (2).
4. Repeat steps 1 to 2 for Top Horizontal Beam (1).

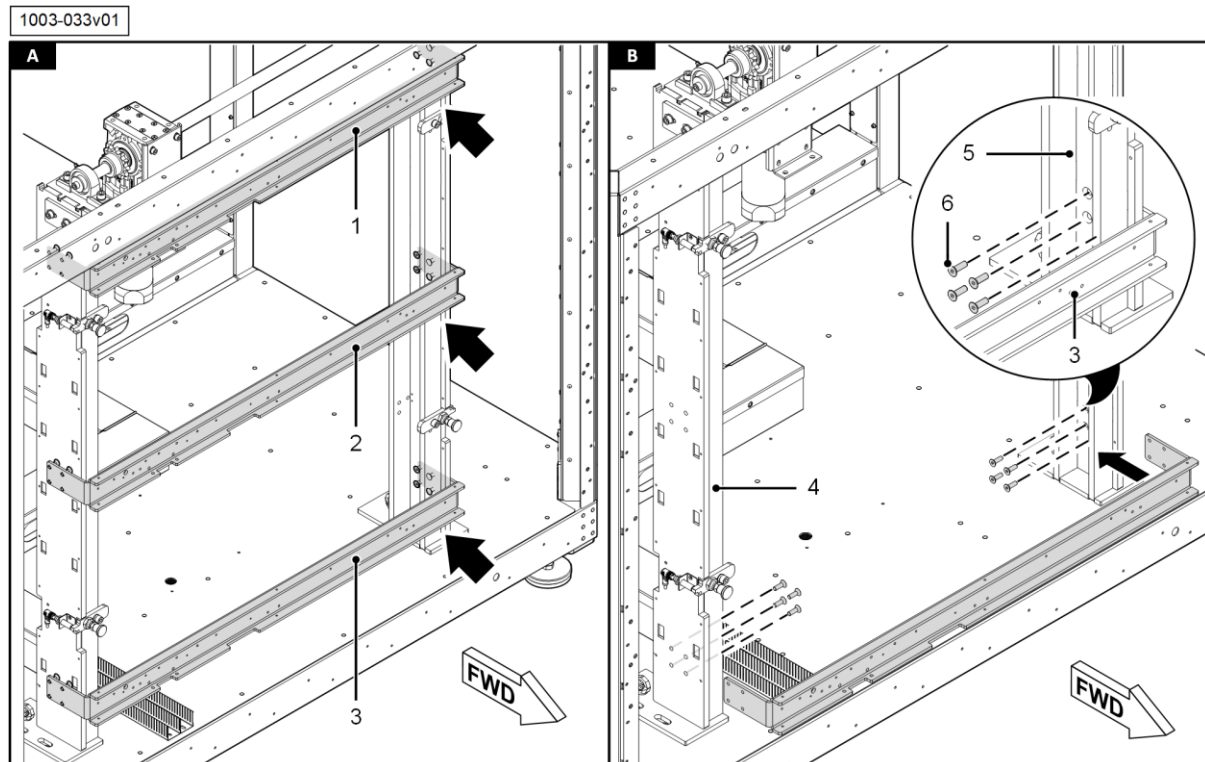


Figure 2-8: Horizontal Beams Installation

2.4.3 Drive Shaft Assembly

2.4.3.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-9	MH230048	Gear	1	Component
2	Figure 2-9	FS230178	DIN985 M6 Nut	5	Fastener
3	Figure 2-9	FS230142	DIN125 M6 Washer	5	Fastener
4	Figure 2-9	FS230388	DIN912 M6x30 Screw	5	Fastener
5	Figure 2-9	OT230012	Motor	1	Component
6	Figure 2-9	FS230389	DIN985 M8 Nut	4	Fastener
7	Figure 2-9	SL230198	Gear Mount	1	Component
8	Figure 2-9	FS230143	DIN125A M8 Washer	4	Fastener
9	Figure 2-9	FS230386	DIN912 M8x25 Screw	4	Fastener
10	Figure 2-9	N/A	Groove – Prat of MH230048	--	N/A
11	Figure 2-9	CC230081	Drive Shaft	1	Component
12	Figure 2-9	MH230045	Feather Key 90mm	1	Fastener
13	Figure 2-9	MH230044	Feather Key 28mm	1	Fastener
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-10	CC230081	Drive Shaft	--	N/A
2	Figure 2-10	CC230082	12B12 Sprocket	1	Component
3	Figure 2-10	FS230090	DIN471 d25 Circlip	1	Fastener
4	Figure 2-10	MH230044	Feather Key 28mm	--	N/A
5	Figure 2-9	N/A	Groove – Prat of CC230082	--	N/A
6	Figure 2-10	MH230047	Housing Bearing	1	Component
7	Figure 2-10	FS230399	DIN9021 M12 Washer	1	Fastener
8	Figure 2-10	FS230374	DIN912 M10x20 Screw	1	Fastener
9	Figure 2-10	FS230374	DIN912 M10x20 Screw	1	Fastener
10	Figure 2-10	FS230399	DIN9021 M12 Washer	1	Fastener
11	Figure 2-10	MH230047	Housing Bearing	1	Component
12	Figure 2-10	FS230090	DIN471 d25 Circlip	1	Fastener
13	Figure 2-10	CC230082	12B12 Sprocket	1	Component
14	Figure 2-10	MH230044	Feather Key 28mm	1	Fastener



2.4.3.2 Assembly Procedure

(Refer to Figure 2-9 below).

1. Position Motor (5) in its place (ensuring the orientation matches the direction indicated as 5*) on the Gear (1).
2. Using a 5 mm Allen-head bit and a 10 mm open-end wrench, install four M6 Allen-head screws (4), four flat washers (3) and four M6 nuts (2) which secure the Motor (5) to the Gear (1).
3. Position Gear Mount (7) in its place on the Drive Shaft Gear (1).
4. Using a 6 mm Allen-head bit and a 6 mm open-end wrench, install four M8 Allen-head screws (9), four flat washers (8) and four M8 nuts (6) which secure Gear Mount (7) to Drive Shaft Gear (1).
5. Install 90 mm Feather Key (12) in its place on the Drive Shaft (11).
6. Insert the Drive Shaft (11) into Drive Shaft Gear (1), make sure to align 90 mm Feather Key (12) with the Drive Shaft groove.
7. Install 28 mm Feather Key (13) in its place on Drive Shaft (11).

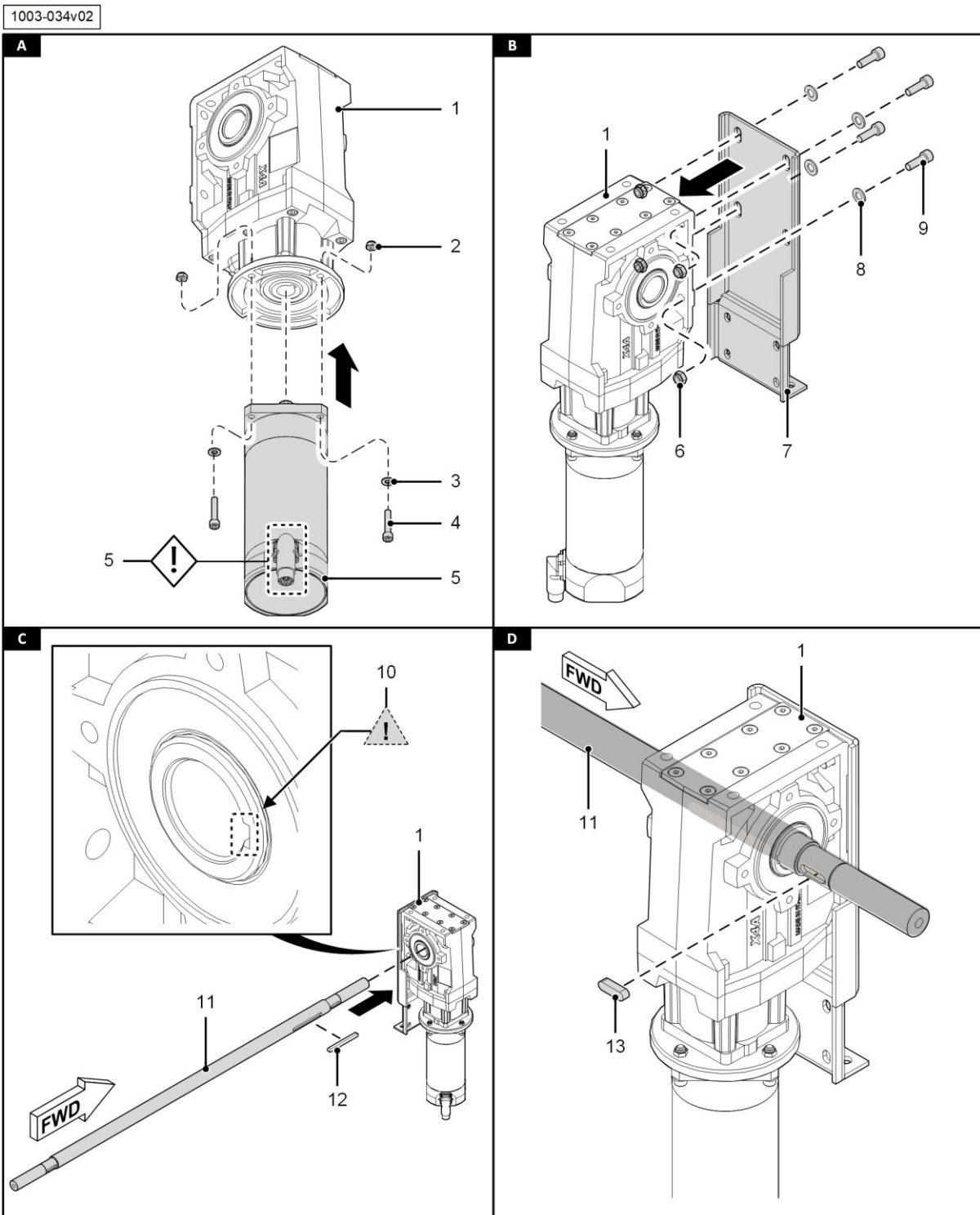


Figure 2-9: Drive Shaft Assembly (1 of 2)

(Refer to Figure 2-10 below).

8. Slide the Sprocket (2) onto Drive Shaft (1), make sure to align 28 mm Feather Key (4) with the Sprocket groove (5).
9. Push the Sprocket (2) as to the end on Drive Shaft (1) so the Drive Shaft groove will be visible.
10. Using circlip pliers, install circlip (3) in its groove on the Drive Shaft (1).
11. Install Housing Bearing (6) onto Drive Shaft (1).
12. Using an 8 mm Allen bit, install M10 Allen-head screw (8) and flat washer (7) which secure the Housing Bearing (6) on the Drive Shaft (1).
13. On the other side of the Drive Shaft, Install 28 mm Feather Key (14) in its place on the Drive Shaft (1).
14. Slide the Sprocket (13) onto Drive Shaft (1), make sure to align 28 mm Feather Key (14) with the Sprocket groove (15).
15. Position the Sprocket (13) as close as possible to Drive Shaft (1) so the Drive Shaft groove will be visible.
16. Using circlip pliers, install circlip (12) in its groove on the Drive Shaft (1).
17. Install Housing Bearing (11) onto Drive Shaft (1).
18. Using an 8 mm Allen bit, install M10 Allen-head screw (9) and flat washer (10) which secure the Housing Bearing (11) on the Drive Shaft (1).

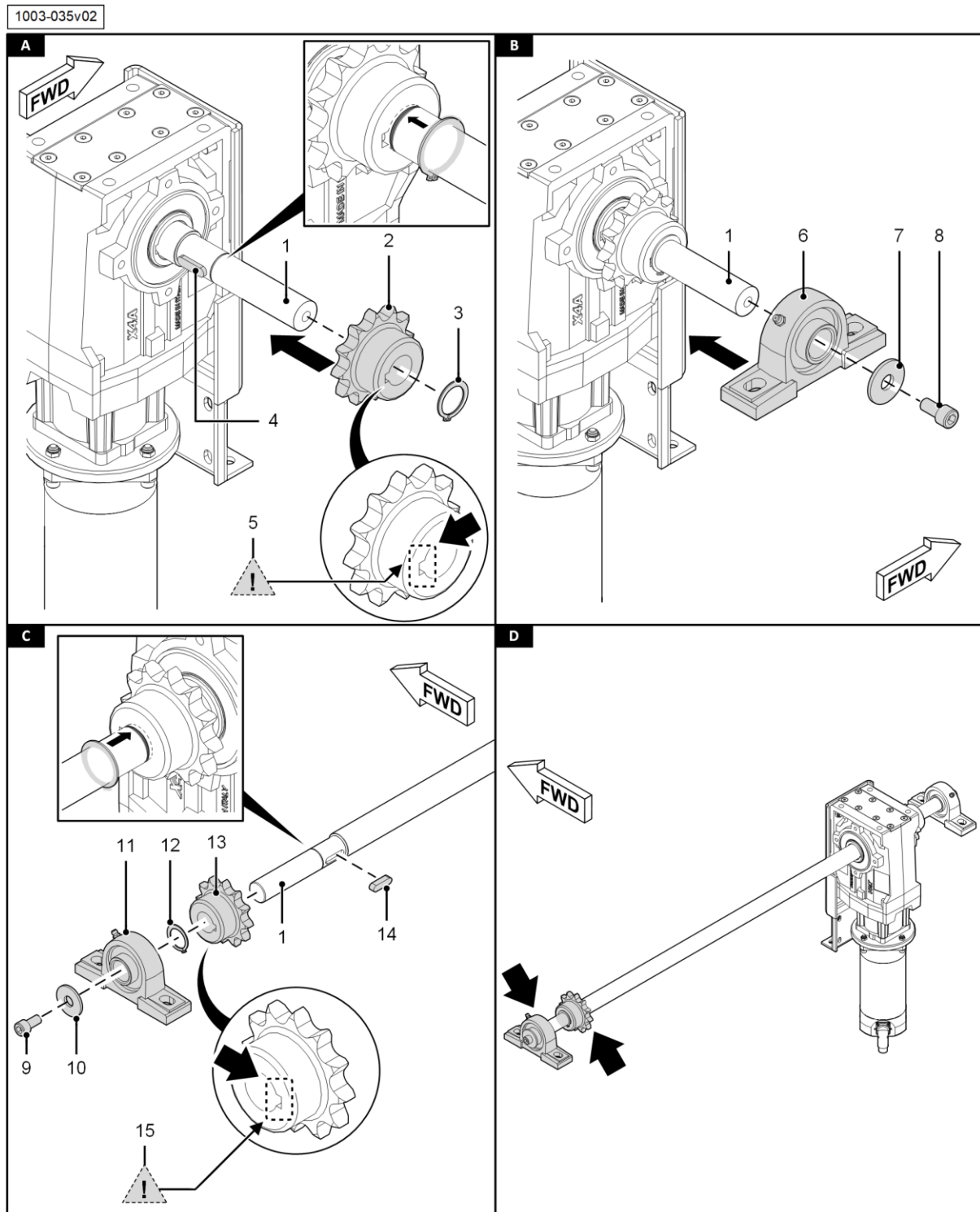


Figure 2-10: Drive Shaft Assembly (2 of 2)

2.4.4 Drive Shaft Installation

2.4.4.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-14	MH230048	Gear	--	N/A
2	Figure 2-14	FS230272	DIN912 M10x30 Screw	4	Fastener
3	Figure 2-14	FS230146	DIN125 M10 Washer	4	Fastener
4	Figure 2-14	AS230134	Right Beam	--	N/A
5	Figure 2-14	AS230136	Left Beam		N/A
6	Figure 2-3	SL230171	Lift Bracket Left	1	Component
7	Figure 2-1	FS230146	DIN125 M10 Washer	4	Fastener
8	Figure 2-1	FS230374	DIN912 M10x20 Screw	4	Fastener
9	Figure 2-3	SL230168	Lift Bracket Right	1	Component
10	Figure 2-14	SL230172	Top Horizontal Beam	--	N/A
11	Figure 2-14	FS230143	DIN125A M8 Washer	6	Fastener
12	Figure 2-14	FS230385	DIN912 M8x16 Screw	6	Fastener

2.4.4.2 Installation Procedure

(Refer to Figure 2-14 below).

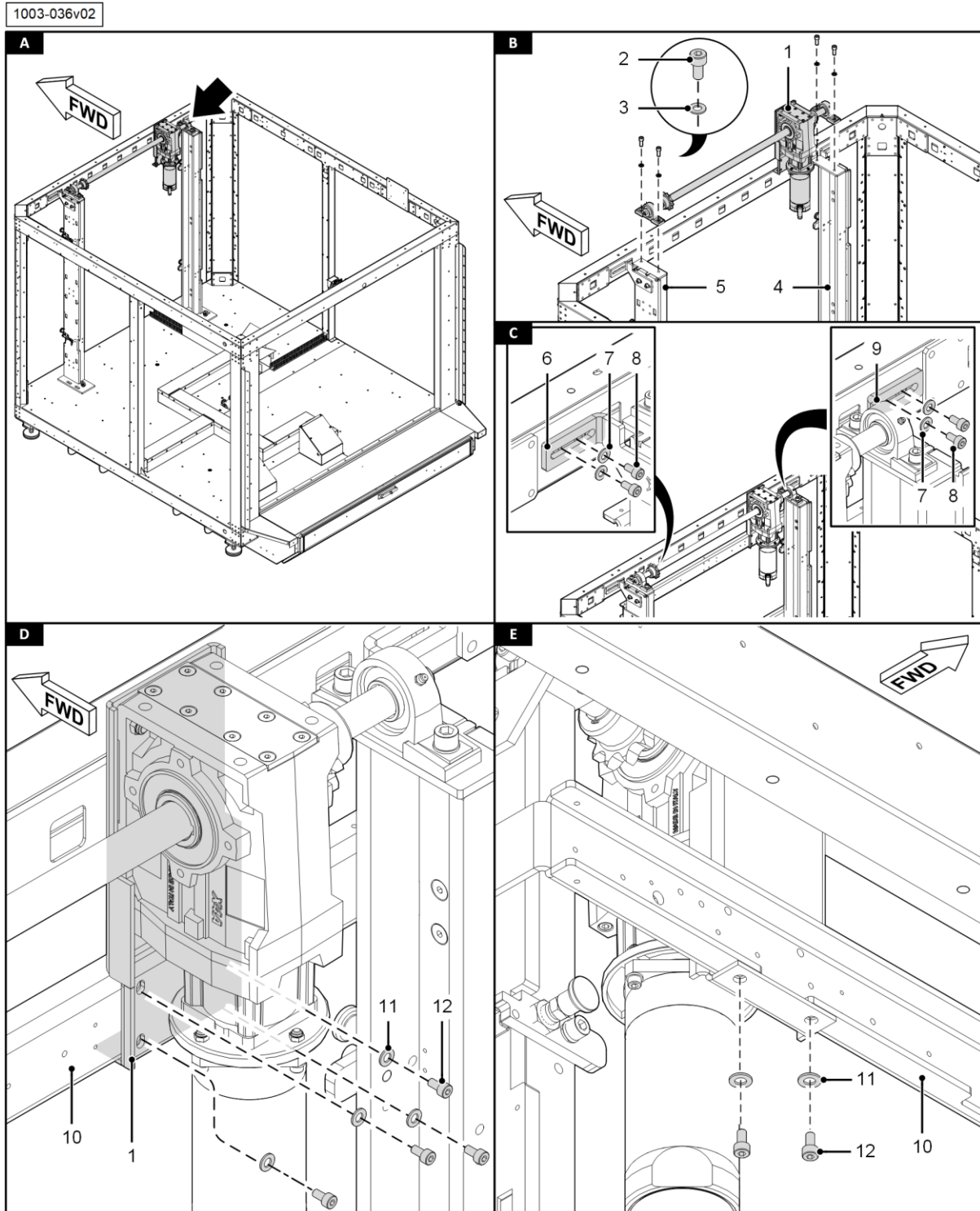
1. Position the Drive Shaft Assembly (1) in its place on both Vertical Rail Beams (4 and 5).

Note

Physically adjust the Right and Left Beams (4 and 5) so the Drive Shaft Assembly installation bores aligned with Right and Left Beams (4 and 5) installation bores.

2. Using an 8 mm Allen bit, install four Allen-head screws (2) and four flat washers (3), which secure the Drive Shaft Assembly (1) to the Vertical Rail Beams (4 and 5).
3. Using an 8 mm Allen bit, install two M10 Allen-head screws (8) and two M10 washers (7), to secure Left Lift Bracket (6) to the UPS and FAN Electrical Assembly.
4. Using an 8 mm Allen bit, install two M10 Allen-head screws (8) and two M10 washers (7), to secure Right Lift Bracket (9) to the UPS and FAN Electrical Assembly.
5. Using an 8 mm Allen bit, install two M10 Allen-head screws (Figure 2-1, 2) and two M10 flat washers (Figure 2-1,3) which secure the Left Beam (Figure 2-1,1) to the UPS and FAN Electrical Assembly.
6. Using an 8 mm Allen bit, install two M10 Allen-head screws (Figure 2-1, 2) and two M10 flat washers (Figure 2-1,3) which secure the Left Beam (Figure 2-1,1) to the UPS and FAN Electrical Assembly.
7. Using an 8 mm Allen bit, install two M10 Allen-head screws (Figure 2-11, 4) and two M10 flat washers (Figure 2-12, 5) which secure the Right Beam (Figure 2-13, 1) to the UPS and FAN Electrical Assembly.

8. Using a 6 mm Allen bit, install six Allen-head screws (12) and six flat washers (11), which secure the Drive Shaft Assembly (1) to the upper Horizontal Beam (10).



2.4.5 Closed Chain Assembly

2.4.5.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-15	FS230377	DIN471 D6 Circlip	1	Fastener
2	Figure 2-15	SL230193	Pulley Mount	1	Component
3	Figure 2-15	CC230078	Idler Shaft	1	Component
4	Figure 2-15	FS230377	DIN471 D6 Circlip	1	Fastener
5	Figure 2-15	CC230077	Chain Idler	1	Components
6	Figure 2-15	SL230172	Bottom Horizontal Beam	--	N/A
7	Figure 2-15	SL230193	Pulley Mount	--	N/A
8	Figure 2-15	FS230141	DIN125 M5 Washer	4	Fastener
9	Figure 2-15	FS230384	DIN912 M5x10 Screw	4	Fastener

2.4.5.2 Assembly Procedure

(Refer to Figure 2-15 below).

Note

The installation bores of the Chain Idler (5) are elliptical shape to provide positioning option – position the Chain Idler (5) in the lower position as possible

1. Position Chain Idler (5) in its place in Pulley Mount (2).
2. Insert Idler Shaft (3) in Pulley Mount (2) and the Chain Idler (5)
3. Using circlip pliers, install two circlips (1 and 4) on the Idler Shaft (3) both ends.
4. Position two Closed Chain Assemblies (7) in their place on the Bottom Horizontal Beam (6).

Note

Install the screws in the center of the oval bores on the Closed Chain Assemblies. Do not apply Loctite.

5. Using a 4 mm Allen-head bit, install eight Allen-head screws (9) and eight flat washers (8), which secure the two Closed Chain Assemblies (7) to the Horizontal Beam (6).



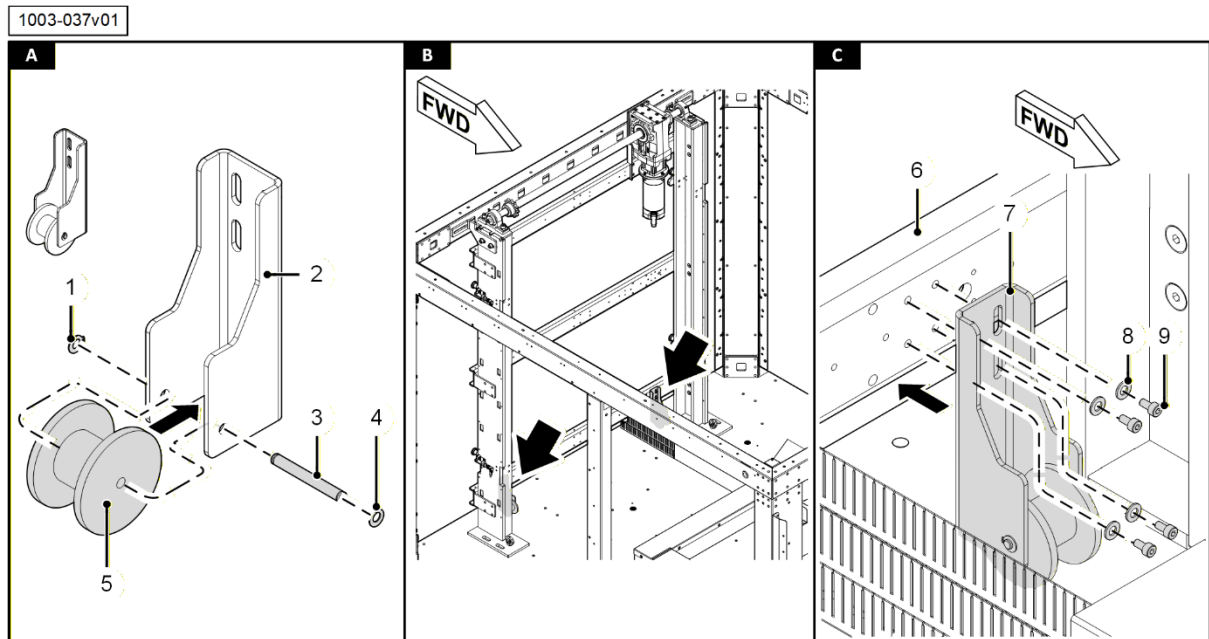


Figure 2-15: Closed Chain Assembly

2.4.6 Roller Chain Adjustment

2.4.6.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-16	MH230046	Roller Chain	1	Component
2	Figure 2-16	Part of MH230046	Pin Assembly	1	--
3	Figure 2-16	MH230091	Spring Clip	1	Fastener
4	Figure 2-16	Part of MH230046	Link Plate	1	--
5	Figure 2-16	CC230080	Chain Centering	1	Component
6	Figure 2-16	MH230049	Swing Bolt M10	2	Fastener
7	Figure 2-16	FS230275	DIN934 M10 Nut	2	Fastener
8	Figure 2-16	FS230146	DIN125 M10 Washer	2	Fastener
9	Figure 2-16	FS230146	DIN125 M10 Washer	2	Fastener
10	Figure 2-16	FS230275	DIN934 M10 Nut	2	Fastener
11	Figure 2-16	Part of MH230046	Connection Pin	--	N/A

2.4.6.2 Adjustment Procedure

(Refer to Figure 2-16 below).

1. Install one M10 nut (7) onto the Swing Bolt (6) and thread the nut down until it reaches the bolt base.
2. Install two flat washers (8 and 9) onto the Swing Bolt (6).
3. Install another M10 nut (10) onto the Swing Bolt (6) and thread the nut close to both flat washers (8 and 9).
4. Connect the Swing Bolt (6) to the Roller Chain (1) as follows:
 - a. Insert a Chain Centering (5) into Swing Bolt (6) bore.
 - b. Insert Pin Assembly (2) through the Swing Bolt (6) bore and the Chain installation bores.
 - c. Install Link Plate (4) on Pin Assembly (2) both pins.
 - d. Install spring Clip (3) on Pin Assembly both pins (2) groove to secure the Swing Bolt (6) to the Roller Chain (1).
5. Fully extend the Roller Chain (1) and measure 2.88 meters from the end, corresponding to **152** pins.
6. Remove the connection pin (11) from Roller Chain **152nd** pin (1).
7. Repeat steps 1 to 4 to install another Swing Bolt (6) on the other end of the Roller Chain (1).

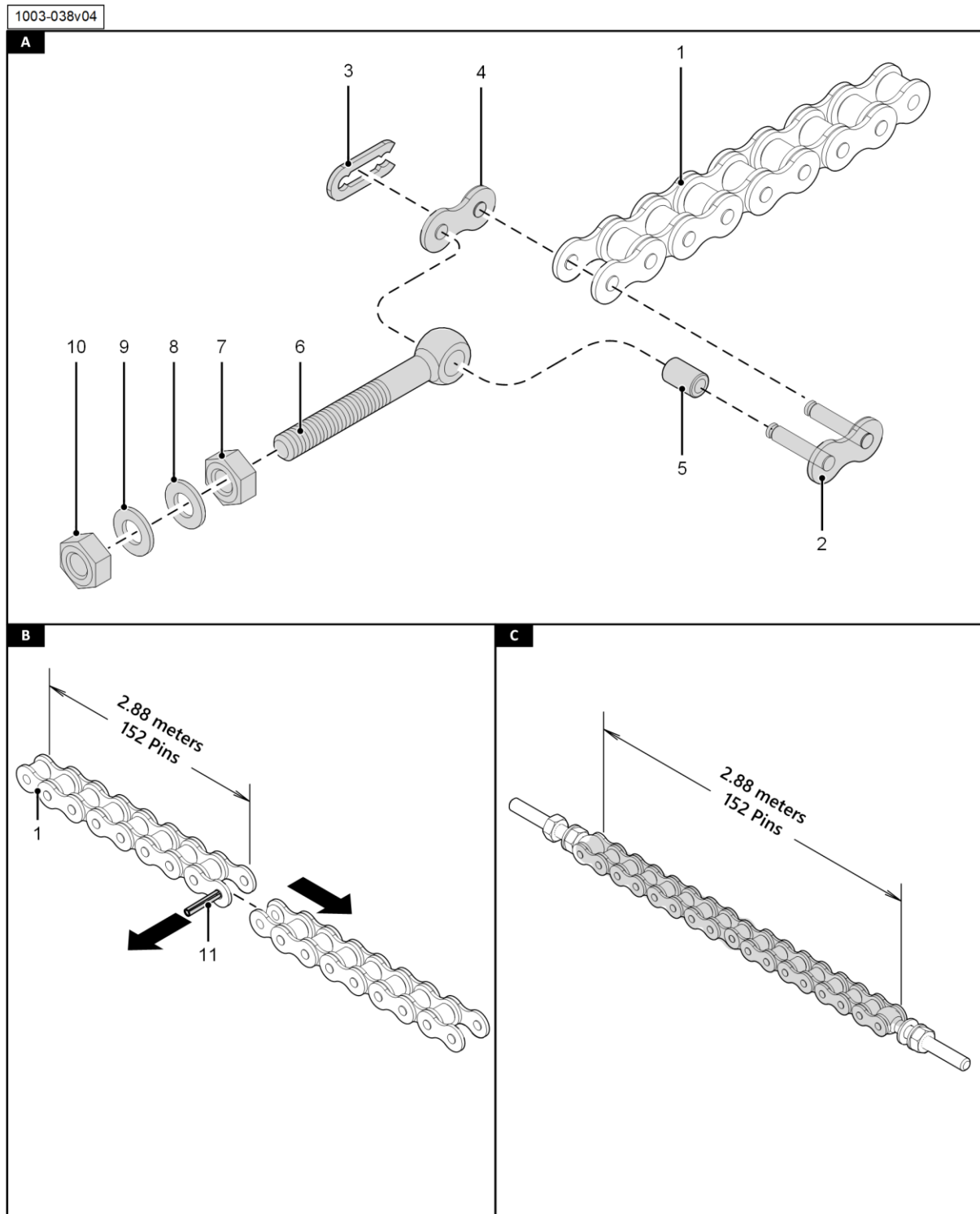


Figure 2-16: Roller Chain Adjustment

2.4.7 Right/Left Roller Assembly

2.4.7.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-17	FS230398	DIN471 d20 Circlip	4	Fastener
2	Figure 2-17	MH230041	Heavy Duty Wheel	4	Component
3	Figure 2-17	CC230075	Wheel Shaft	4	Component
4	Figure 2-17	SL230174	Left Roller Base	1	Component
		SL230184	Right Roller Base	1	Component
5	Figure 2-17	FS230144	DIN125 M16 Washer	4	Fastener
6	Figure 2-17	FS230379	DIN985 M16 Nut	4	Fastener
7	Figure 2-17	FS230377	DIN471 D6 Circlip	4	Fastener
8	Figure 2-17	SL230173	Side Wheel Bracket	2	Component
9	Figure 2-17	CC230076	Side Wheel Shaft	2	Component
10	Figure 2-17	MH230042	Wheel	2	Component
11	Figure 2-17	FS230377	DIN471 D6 Circlip	4	Fastener
12	Figure 2-22	FS230284	mounting pin - DIN6325 D8x34	4	Fastener
13	Figure 2-17	FS230148	DIN125A M6 Washer	4	Fastener
14	Figure 2-17	CC230076	Side Wheel Shaft	2	Component
15	Figure 2-17	FS230380	DIN933 M6x35 Screw	2	Component
16	Figure 2-17	FS230321	DIN934 M6 Nut	2	Fastener
17	Figure 2-17	SL230178	Tension Screw Holder	2	Component
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-18	FS230220	DIN912 M5x12 Screw	4	Fastener
2	Figure 2-18	FS230141	DIN125 M5 Washer	4	Fastener
3	Figure 2-18	SL230181	Lift Flag	2	Component
4	Figure 2-18	SL230183	Lift Flag Base	1	Component
5	Figure 2-18	SL230182	Lift Flag Nut	2	Fastener
6	Figure 2-18	SL230180	Right Roller	1	Component
7	Figure 2-18	FS230220	DIN912 M5x12 Screw	4	Fastener
8	Figure 2-18	FS230141	DIN125 M5 Washer	4	Fastener
9	Figure 2-18	FS230160	Nut	4	Fastener



2.4.7.2 Assembly Procedure

(Refer to Figure 2-17 below).

1. Perform Left Roller assembly as follows:
 - a. Insert two Wheel Shafts (3) into Roller Base installation bores (4).
 - b. Using a 24 mm open-end wrench, install two M16 nuts (6) and two M16 flat washers (5) which secure Wheel Shafts (3) to the Roller Base (4).
 - c. Slide two Wheels (2) onto Wheel Shafts (3) until fully seated.
 - d. Using circlip pliers, install one circlip (1) in its groove on each Wheel Shaft (3).
 - e. Position the Side Wheel (10) at the center of the Side Wheel Bracket (8), aligned between the shaft installation bores.
 - f. Insert Side Wheel Shaft (9) into Side Wheel Bracket (8) through the Side Wheel (10), make sure the Side Wheel Shaft (9) is positioned at the center.
 - g. Using circlip pliers, install two circlips (7) into the groove on each Side Wheel Shaft (9) to secure the Side Wheel (10) and the shaft to the Side Wheel bracket (8).
 - h. Position the Side Wheel Bracket (8) between two Shaft Holders, aligned between the Shaft Holders and Side Wheel Bracket installation bores.
 - i. Insert Side Wheel Bracket Shaft (14) into Shaft Holders through the Side Wheel Bracket (8) with two M6 flat washers (13), make sure the Side Wheel Bracket Shaft (14) is positioned at the center.
 - j. Using circlip pliers, install two circlips (11) into the groove on each Side Wheel Shaft (14) to secure the Side Wheel Bracket (8) and the shaft to the Side Wheel Bracket Shaft (14).
 - k. Manually thread M6 nut (16) on M6 Tension Screw (15) to the end its length.
 - l. Manually install M6 Tension Screw (15) into Tension Screw Holder (17) until M6 nut (16) touches Tension Screw Holder (17).
 - m. Using Plastic Hammer install two mounting pins (12) into four pin bores.

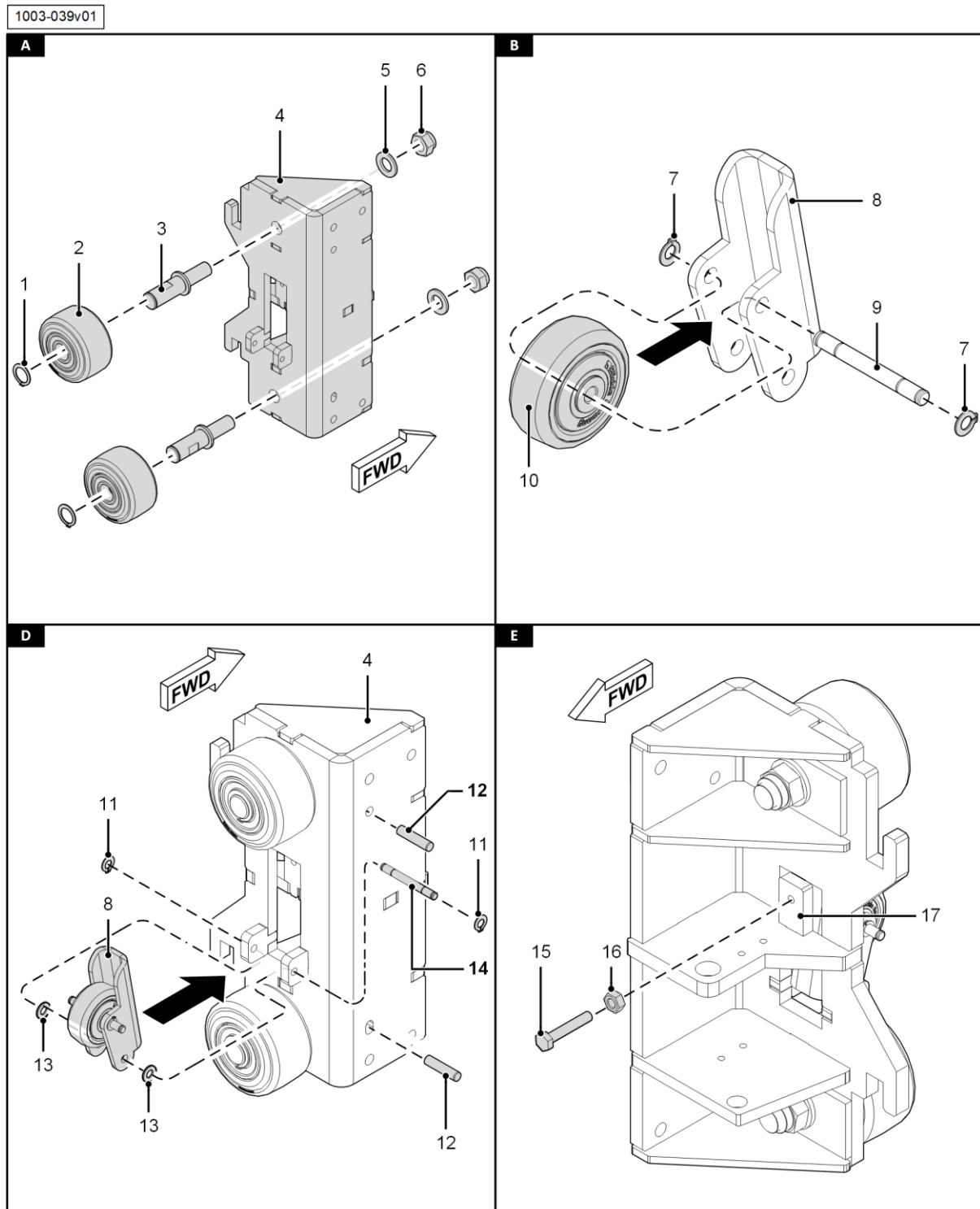


Figure 2-17: Right/Left Roller Assembly

2. Repeat steps a to m for Right Roller assembly.

Note

Perform the following steps only for the Right Roller Assembly

(Refer to Figure 2-18 below).

3. Perform Lift Flag assembly as follows:
 - a. Position two Lift Flags (3 and 6) on Lift Flag Base (4).
 - b. Using a 4 mm Allen-head bit, install four (two on each Lift Flag) Allen-head screws (1), four flat washers (2) and two Lift Flag Nuts (5), which secure the two Lift Flags (3 and 6) to Lift Flag Base (4).
4. Position Lift Flag assembly (4) in its place on Right Roller assembly (7).
5. Using a 4 mm Allen-head bit, install four Allen-head screws (8) and four flat washers (9), which secure the Lift Flag assembly (4) to Right Roller assembly (7).

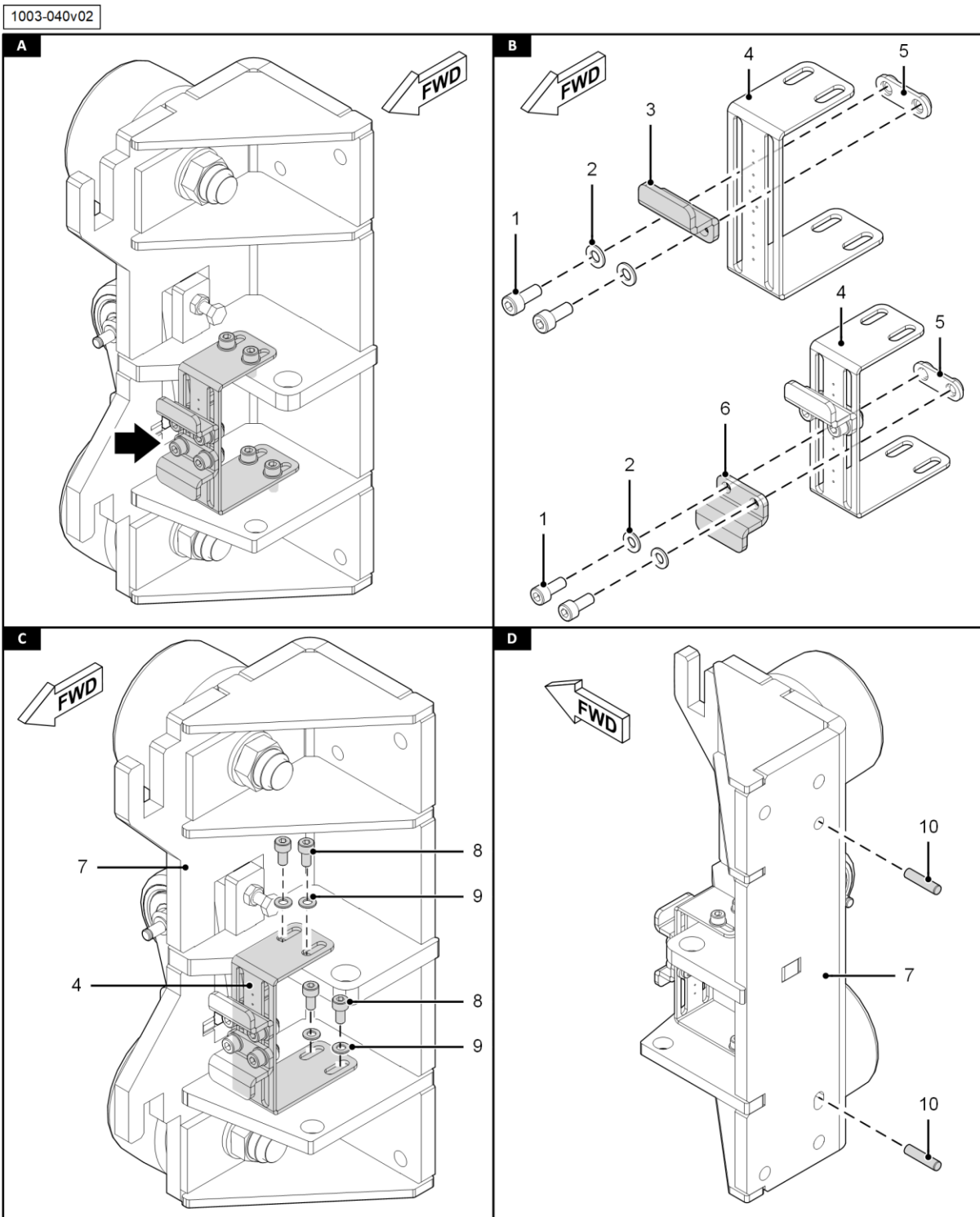


Figure 2-18: Lift Flag assembly

2.4.8 Right/Left Roller Installation

2.4.8.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-19	AS230134	Right Beam	--	Component
2	Figure 2-19	AS230147	Right Roller	--	Component
3	Figure 2-19	AS230146	Left Roller	--	Component
4	Figure 2-19	AS230136	Left Beam	--	Component

2.4.8.2



2.4.8.3 Installation Procedure

(Refer to Figure 2-19 below).

1. Position Right Roller Assembly (2), Left Roller Assembly (3) so that their wheels are located inside the right (1) and left (4) vertical rail beams.

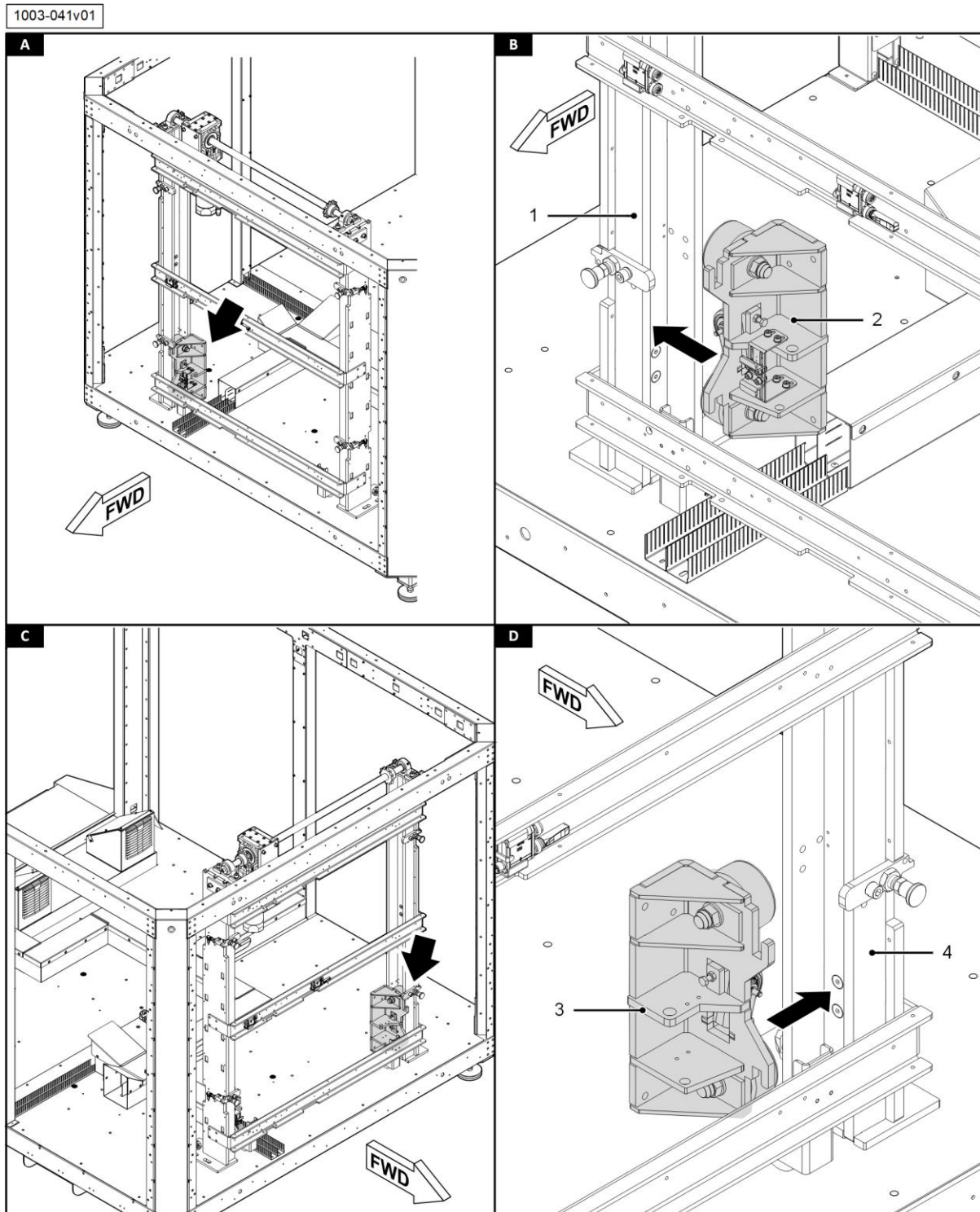


Figure 2-19: Right/Left Roller Installation

2.4.9 Cart Assembly

2.4.9.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-20	AS230149	Lift Electric Box	1	Component
2	Figure 2-20	Part of AS230149	Philips-head screws	6	Component
3	Figure 2-20	Part of AS230149	Lift Electric Box Cover	1	Fastener
4	Figure 2-20	AS230140	Cart Assembly	1	Component
5	Figure 2-20	FS230278	DIN912 M4x8 Screw	4	Fastener
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-21	SL230187	Igus Chain Bracket	-	Component
2	Figure 2-21	SL230190	Igus U Bracket	1	Component
3	Figure 2-21	FS230141	DIN125 M5 Washer	4	Fastener
4	Figure 2-21	FS230384	DIN912 M5x10 Screw	4	Fastener
5	Figure 2-21	SL230186	Feet Bracket	-	Component
6	Figure 2-21	Part of MH230043	Nut	2	Fastener
7	Figure 2-21	MH230043	Leveling Foot	2	Component
8	Figure 2-21	LS230021	NIL Ribbed Tube End Plug	4	Component



2.4.9.2 *Installation Procedure*

(Refer to Figure 2-20 below).

1. Install Lift Electric Box (1) as follows:
 - a. Using a Philips-head bit, remove six Philips-head screws (2) which secure Lift Electric Box Cover (3) to the Lift Electric Box (1).
 - b. Remove Lift Electric Box Cover (3) from the Lift Electric Box (1).
 - c. Position Lift Electric Box (1) on the Electrical Box Holder (4) which welded to the Cart Assembly.
 - d. Using a 3 mm Allen-head bit, install four M4 Allen-head screws (5) which secure Lift Electric Box (1) to the Electrical Box Holder (4).
 - e. Position Lift Electric Box Cover (3) in its place on the Lift Electric Box (1).
 - f. Using a Philips-head bit, install six Philips-head screws (2) which secure Lift Electric Box Cover (3) to the Lift Electric Box (1).

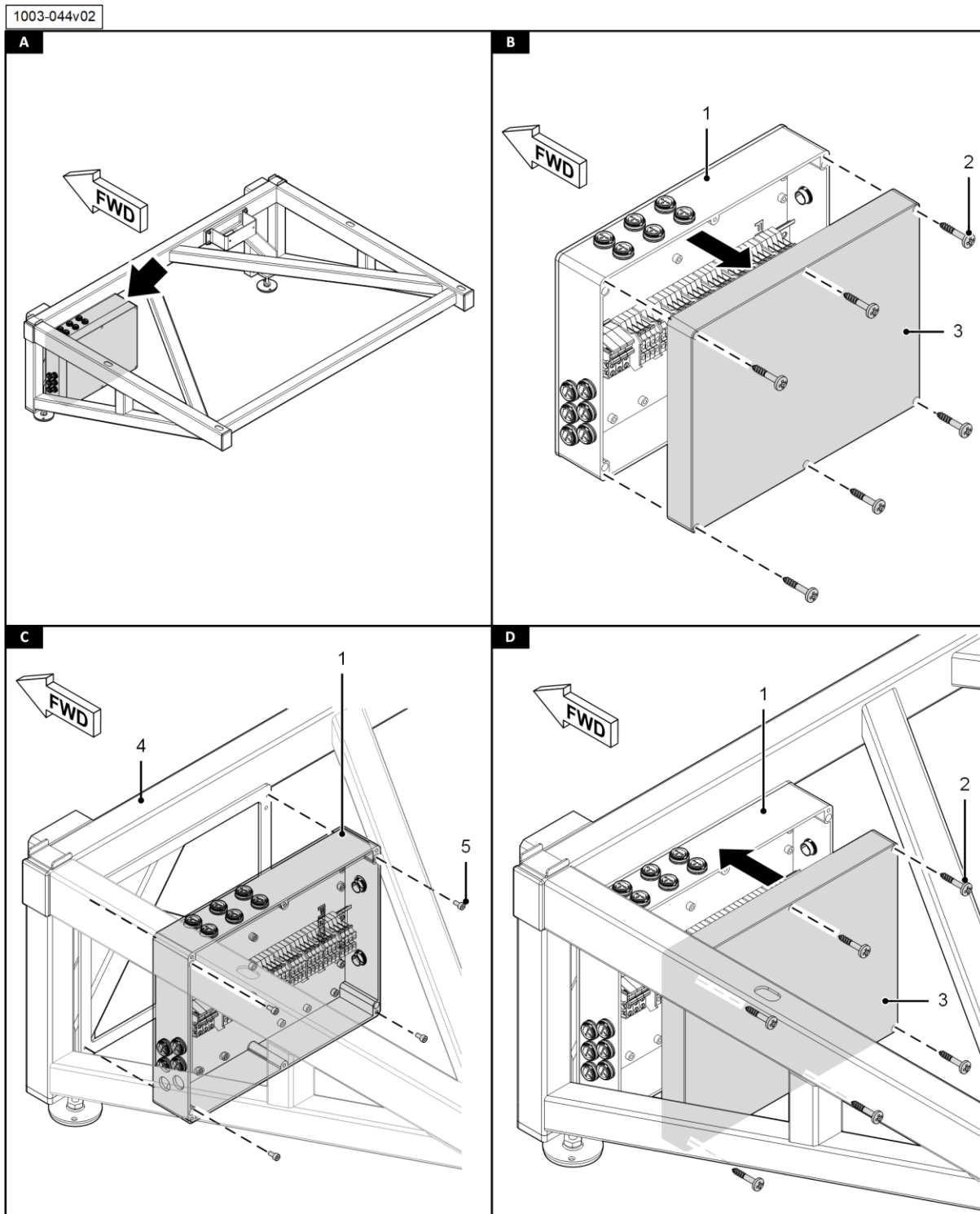


Figure 2-20: Lift Electric Box Installation

(Refer to Figure 2-21 below).

2. Install Igus U Bracket (2) as follows:
 - a. Position Igus U Bracket (2) in its place on the Chain Bracket (1).
 - b. Using a 4 mm Allen-head bit, install four M5 Allen-head screws (4) and four M5 flat washers (3) which secure Igus Bracket (2) to the Chain Bracket (1).
3. Insert two Cart Leveling Foots (7) into Feet Bracket (5) installation bores.
4. Using a 13 mm open-end wrench, release (CCW) the M8 Leveling Nut (6) and fully tighten Cart Leveling Foot (7).
5. Using a 13 mm open-end wrench, tighten (CCW) the M8 Leveling Nut (6).
6. Install four End-Plugs (8) into four Cart corners.

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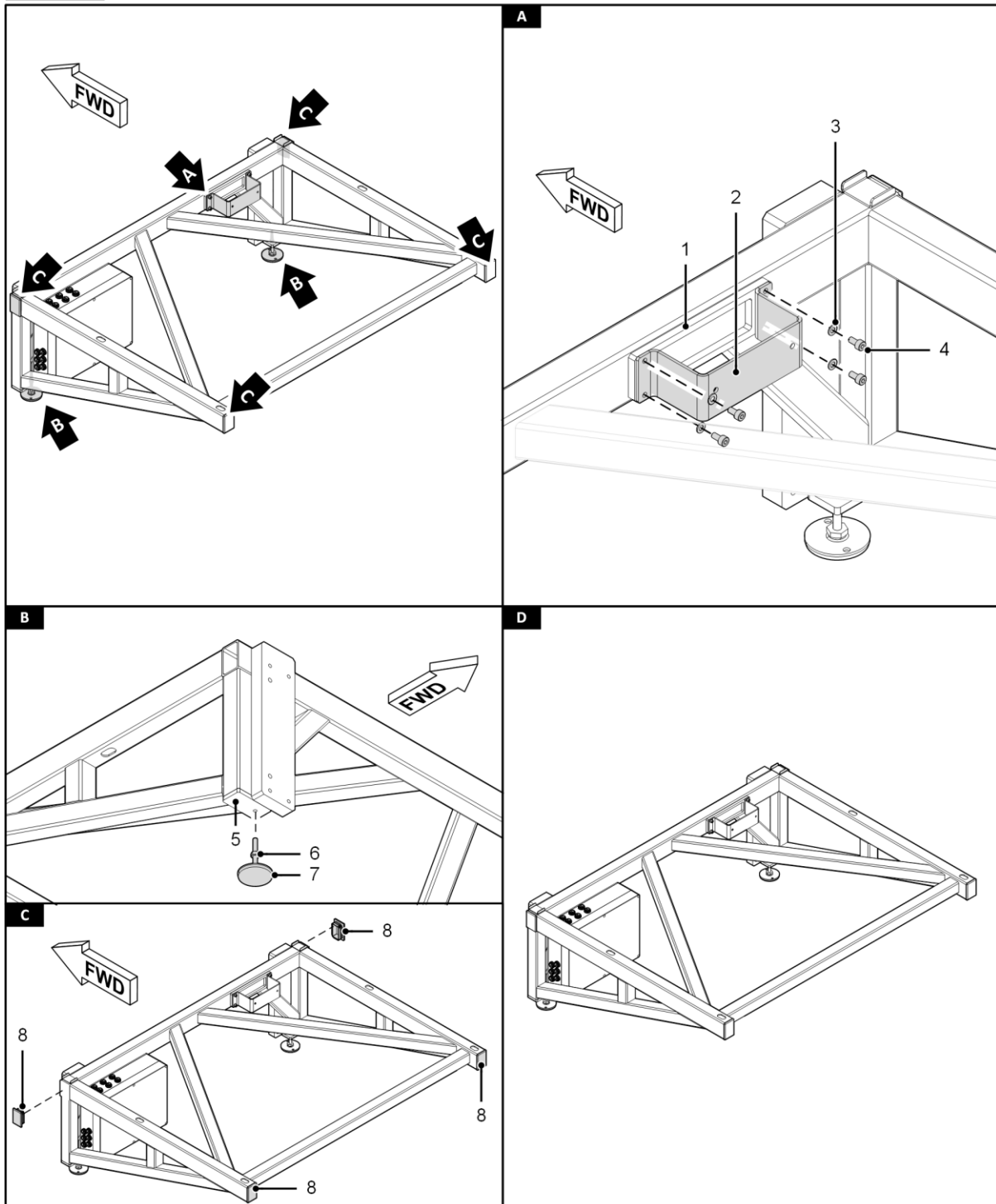


Figure 2-21: Cart Assembly

2.4.10 Cart Installation

2.4.10.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-22	AS230147 AS230146	Right Roller Left Roller	1	Component
2	Figure 2-22	FS230284	mounting pin - DIN6325 D8x34	4	Fastener
3	Figure 2-22	AS230140	Cart Assembly	1	Component
4	Figure 2-22	FS230143	DIN125 M8 Washer	-	Fastener
5	Figure 2-22	FS230386	DIN912 M8x25 Screw	4	Fastener

2.4.10.2 Installation Procedure

(Refer to Figure 2-22 below).

1. Mount the Cart Assembly (3) on Right and Left Rollers (1), make sure that four mounting pins (2) are inserted into four bores on Right and Left Rollers (1).
2. Using a 6 mm Allen-head bit, install eight M8 Allen-head screws (5) and eight M8 flat washers (4) which secure Cart Assembly (3) to Right and Left Rollers (1).



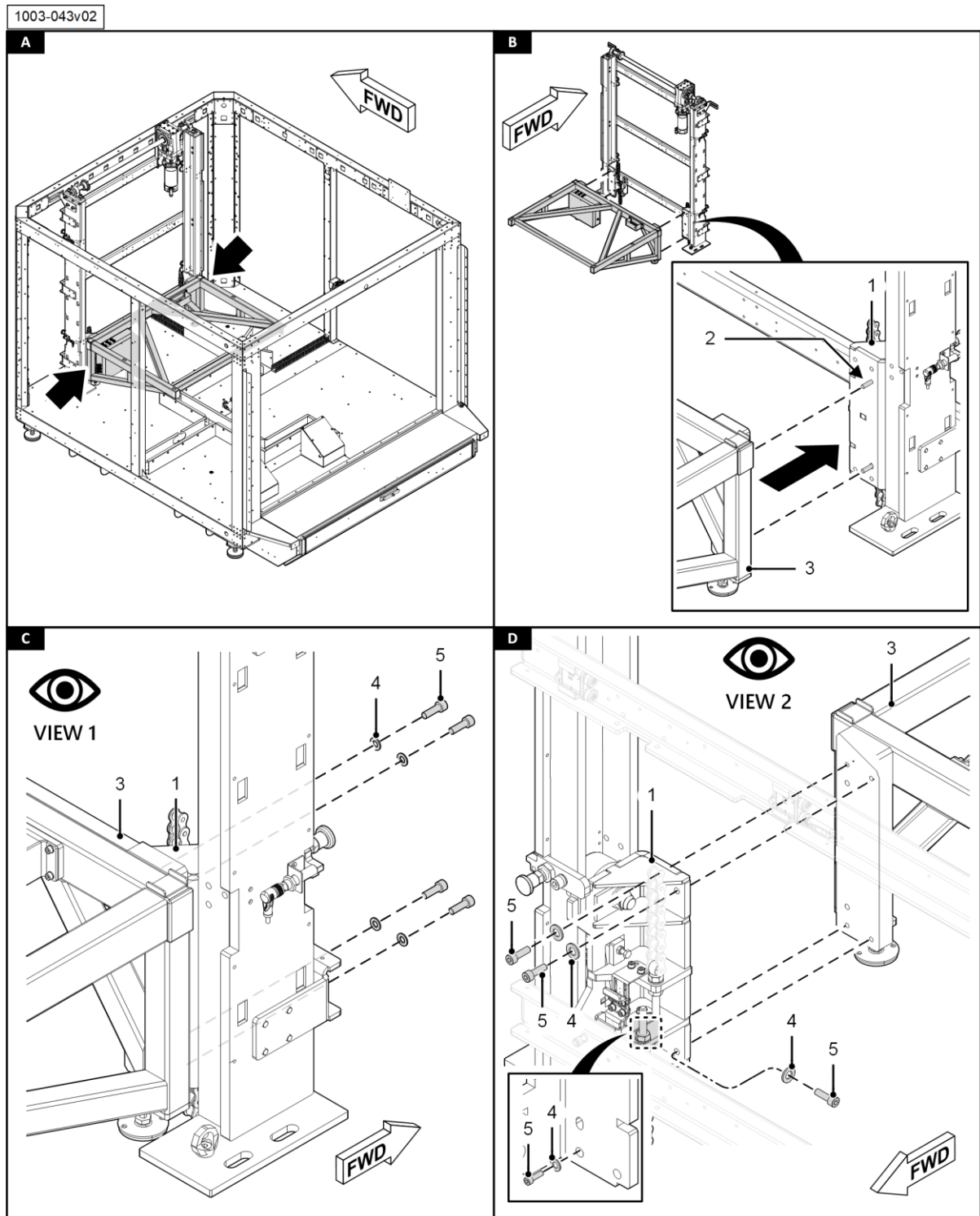


Figure 2-22: Cart Installation

2.4.11 Right/Left Roller Chain Installation

2.4.11.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-23	MH230049	Swing Bolt M10	--	Fastener
2	Figure 2-23	AS230147 AS230146	Right Roller Left Roller	--	Component
3	Figure 2-23	FS230146	DIN125 M10 Washer	--	Fastener
4	Figure 2-23	FS230275	DIN934 M10 Nut	--	Fastener
5	Figure 2-23	CC230082	Sprocket	--	Component
6	Figure 2-23	MH230046	Roller Chain	--	Component
7	Figure 2-23	AS230143	Closed Chain Assembly	--	Component
8	Figure 2-23	CC230077	Idler	--	Component
9	Figure 2-23	MH230049	Swing Bolt M10	--	Fastener
10	Figure 2-23	AS230147 AS230146	Right Roller Left Roller	--	Component
11	Figure 2-23	FS230146	DIN125 M10 Washer	--	Fastener
12	Figure 2-23	FS230275	DIN934 M10 Nut	--	Fastener

2.4.11.2 Installation Procedure

(Refer to Figure 2-23 below).

1. Install Left Roller Chain as follows:
 - a. Remove one M10 nut (4) and one flat washer (3) from the Swing Bolt (1).
 - b. Insert the Swing Bolt (1) into Roller Primary Rib (2) installation bore.
 - c. Install one M10 nut (4) and one flat washer (3) onto the Swing Bolt (1) and thread the nut down until it reaches the Roller Primary Rib (2).
 - d. Route the Roller Chain (6) upwards to the Drive Shaft Sprocket (5).
 - e. Install the Roller Chain (6) on the Drive Shaft Sprocket (5).
 - f. Apply tension to the Roller Chain (6) and position it so the links fully engage the Drive Shaft Sprocket (5) teeth encircling it.
 - g. Ensure the Roller Chain (6) is properly seated and aligned on the Drive Shaft Sprocket (5).
 - h. Route the Roller Chain (6) downward through the Closed Chain Assembly (7) behind the chain Idler (8).
 - i. Remove one M10 nut (12) and one flat washer (11) from the Swing Bolt (9).
 - j. Insert the Swing Bolt (9) into Lower Roller Rib (10) installation bore.
 - k. Install M10 nut (12) and flat washer (11) onto the Swing Bolt (9) and thread the nut down until it reaches the Lower Roller Rib (10).
2. Repeat steps 1.a to step 1.k to install the Right Roller Chain.
3. Apply Loctite LB8101 on both Roller Chains.
4. Position Right and Left Rollers at the lower position by operating the Roller Chains.

5. Using a 4 mm Allen bit, tighten eight Allen-head screws (Figure 2-15, 9) and eight flat washers (Figure 2-15, 8), which secure the two Closed Chain Assemblies to the Horizontal Beam.

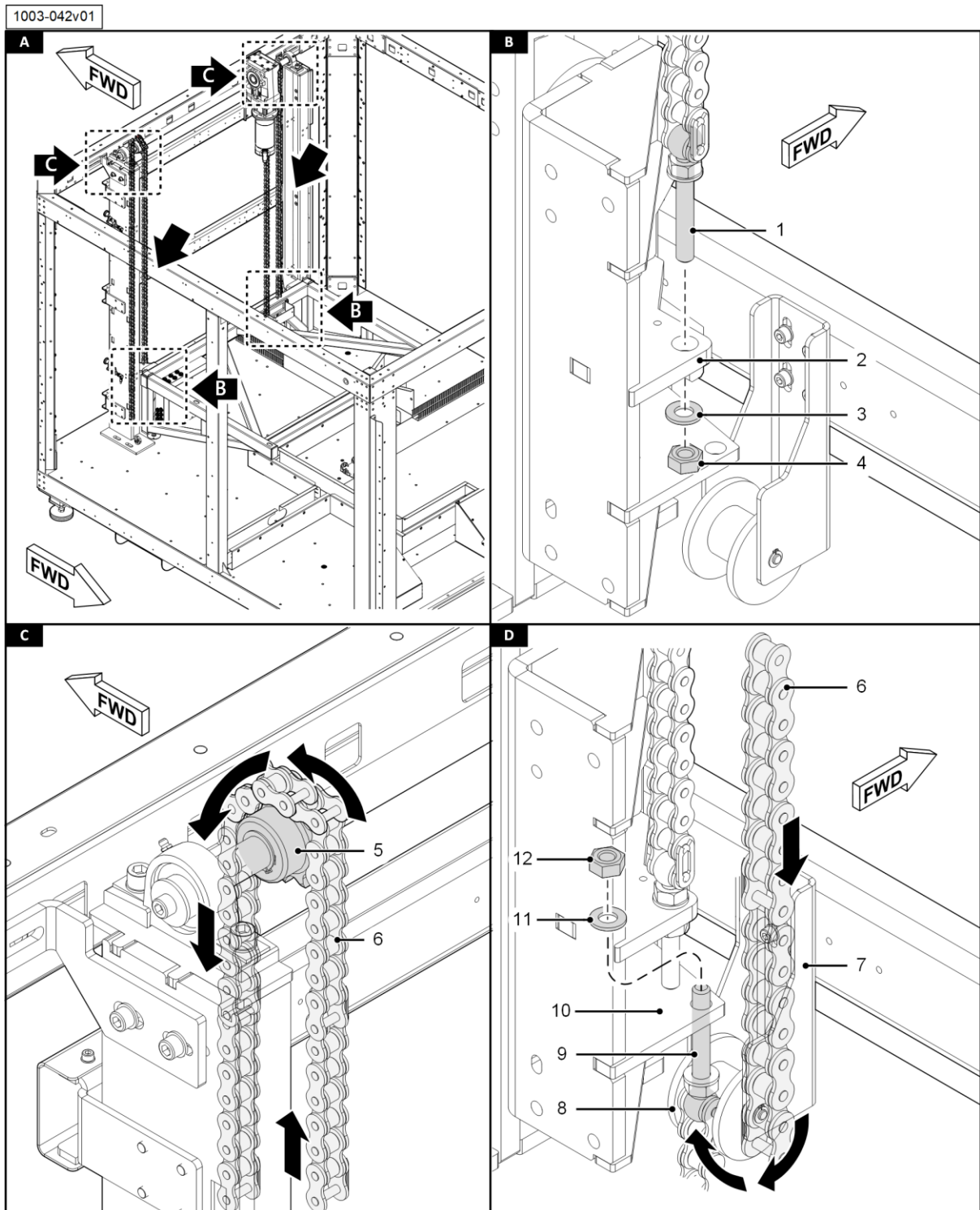


Figure 2-23: Right/Left Roller Chain Installation

2.4.12 Cable Chain (Igus) and Vertical Wire Chanel

2.4.12.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-24	SL230192	Vertical Wire Channel	1	Component
2	Figure 2-24	LS230066	Cable Tie Holder	7	Component
3	Figure 2-24	FS230586	ISO7380 M5X8 Screw	7	Fastener
4	Figure 2-24	SL230172	Middle Horizontal beam	-	Component
5	Figure 2-24	SL230172	Bottom Horizontal beam	-	Component
6	Figure 2-24	FS230142	DIN125 M6 Washer	4	Fastener
7	Figure 2-24	FS230383	DIN912 M6x12 Screw	4	Fastener
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-25	SL230172	Middle Horizontal beam	-	Component
2	Figure 2-25	SL230172	Bottom Horizontal beam	-	Component
3	Figure 2-25	SL230191	Igus Bracket	1	Component
4	Figure 2-25	FS230142	DIN125 M6 Washer	4	Fastener
5	Figure 2-25	FS230383	DIN912 M6x12 Screw	4	Fastener
#	Figure No.	P/N	Description	Qty.	Type
1	Figure 2-26	FS230212	DIN985 M5 Nut	2	Fastener
2	Figure 2-26	FS230141	DIN125 M5 Washer	2	Fastener
3	Figure 2-26	SL230191	Igus Bracket	1	Component
4	Figure 2-26	IG230001	Cable Chain (Igus)	1	Component
5	Figure 2-26	FS230199	DIN912 M5x16 Screw	2	Fastener
6	Figure 2-26	FS230212	DIN985 M5 Nut	2	Fastener
7	Figure 2-26	FS230141	DIN125 M5 Washer	2	Fastener
8	Figure 2-26	SL230190	Igus U Bracket	-	Component
9	Figure 2-26	IG230001	Cable Chain (Igus)	-	Component
10	Figure 2-26	FS230199	DIN912 M5x16 Screw	2	Fastener

2.4.12.2 Installation Procedure

(Refer to Figure 2-24 below).

1. Install Vertical Wire Channel (1) as follows:
 - a. Position seven Tie Mounts (2) in its place on the Vertical Wire Channel (1).
 - b. Using a 4 mm Allen-head bit, install seven M5 Allen-head screws (3) which secure seven Tie Mounts (2) to the Vertical Wire Channel (1).
 - c. Position Vertical Wire Channel (1) in its place on the Middle (4) and Bottom (5) Horizontal Beams.
 - d. Using a 5 mm Allen-head bit, install four M6 Allen-head screws (7) and four M6 flat washers (6) which secure Vertical Wire Channel (1) to the Middle (4) and Bottom (5) Horizontal Beams.

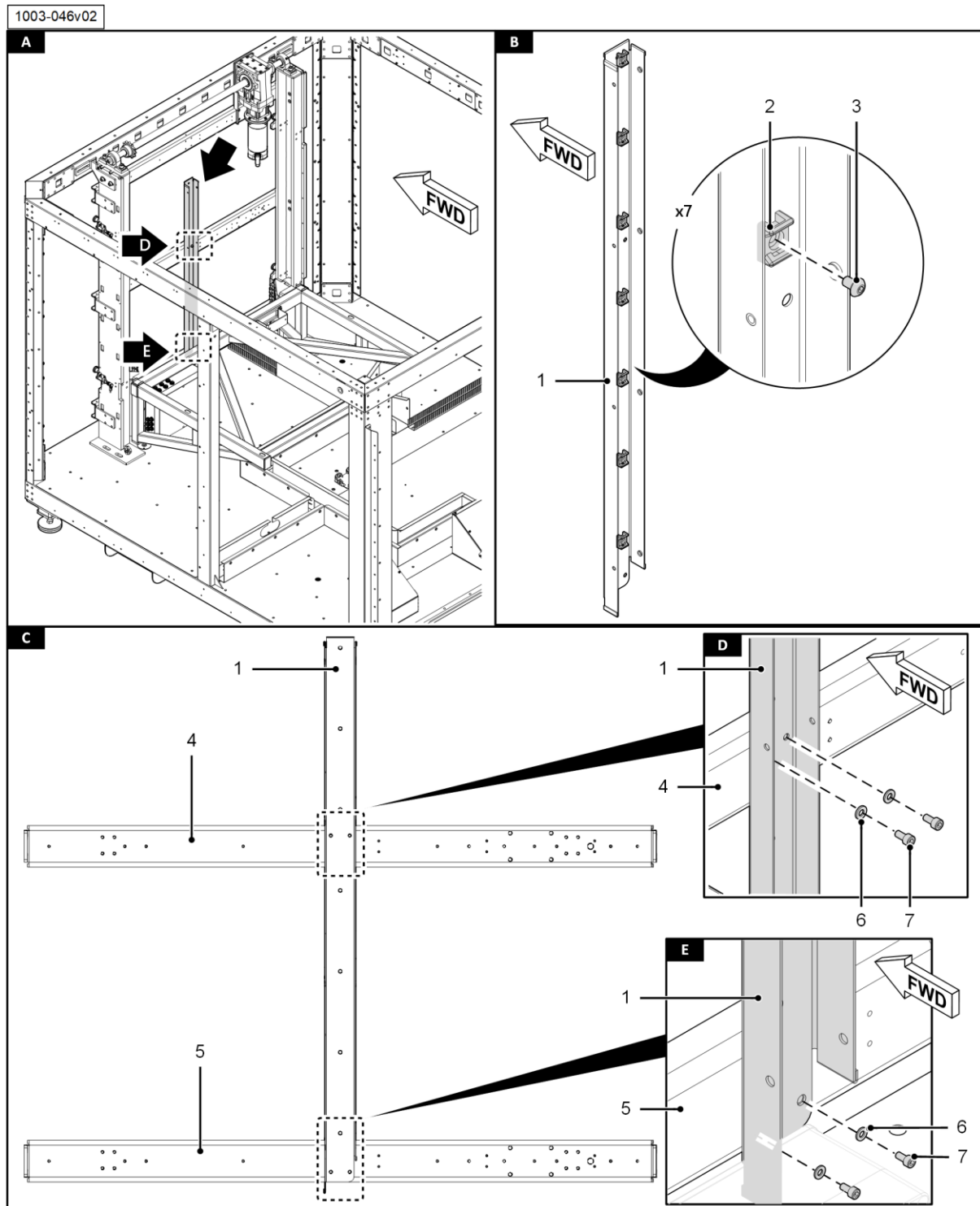


Figure 2-24: Vertical Wire Channel Installation

(Refer to Figure 2-25 below).

2. Install Igus Bracket (3) as follows:
 - a. Position Igus Bracket (3) in its place on the Middle (1) and Bottom (2) Horizontal Beams.
 - b. Using a 5 mm Allen-head bit, install four Allen-head screws (5) and four flat washers (3) which secure Igus Bracket (1) to the Middle (2) and Bottom (5) Horizontal Beams.

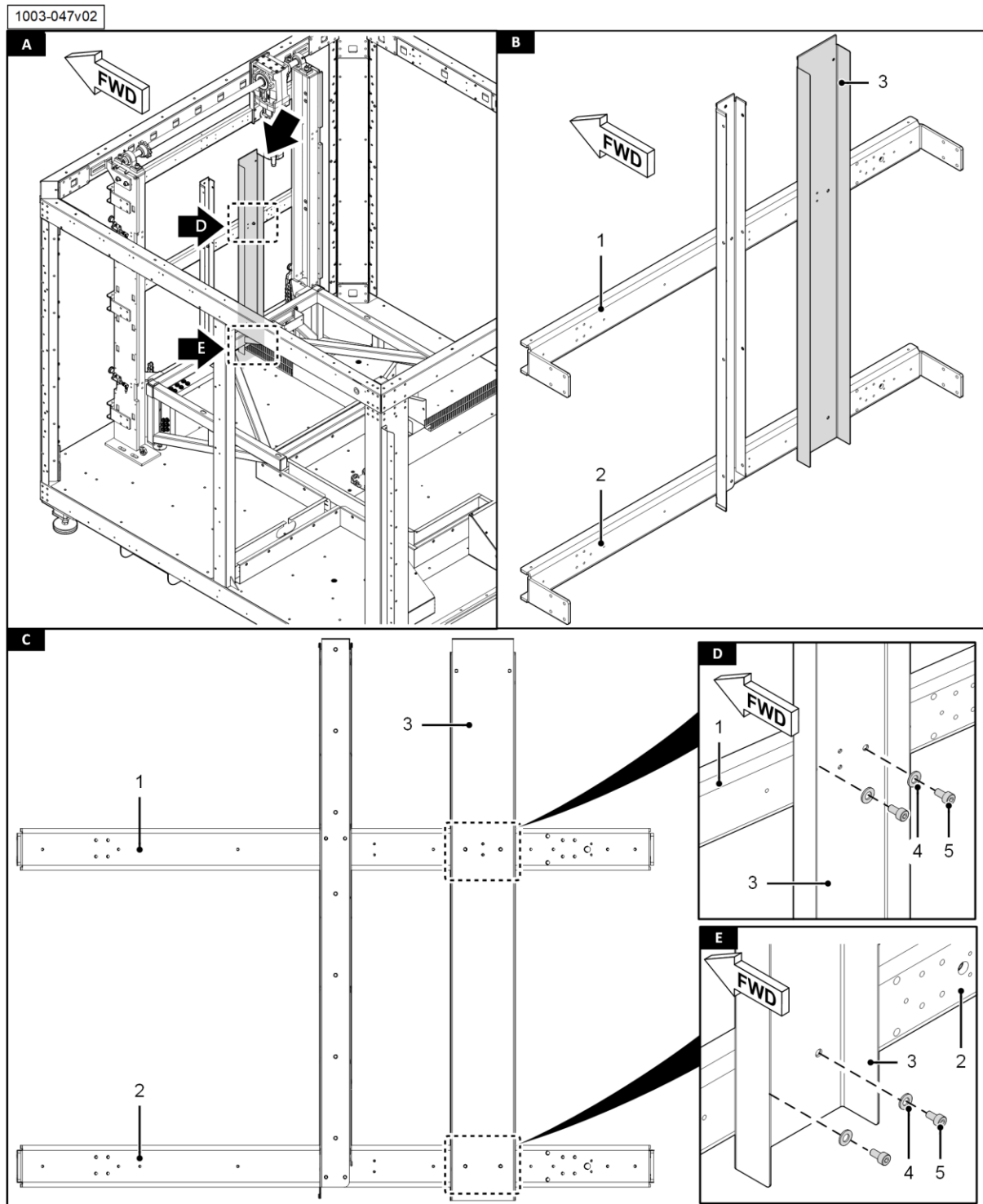


Figure 2-25: Icus Bracket Installation

(Refer to Figure 2-26 below).

3. Install Cable Chain (Igus) (4) as follows:
 - a. Position one side of the Igus (4) in its place on the Igus Bracket (3).
 - b. Using a 4 mm Allen-head bit and an 8 mm open-end wrench, install two Allen-head screws (5), two flat washers (2) and two nuts (1) which secure Igus (4) to the Igus Bracket (3).
 - c. Route the other end Igus and position Igus part on the outer side of the Igus Bracket (8).
 - d. Using a 4 mm Allen-head bit and an 8 mm open-end wrench, install two Allen-head screws (10), two flat washers (7) and two nuts (6) which secure Igus (9) to the Igus Bracket (8).

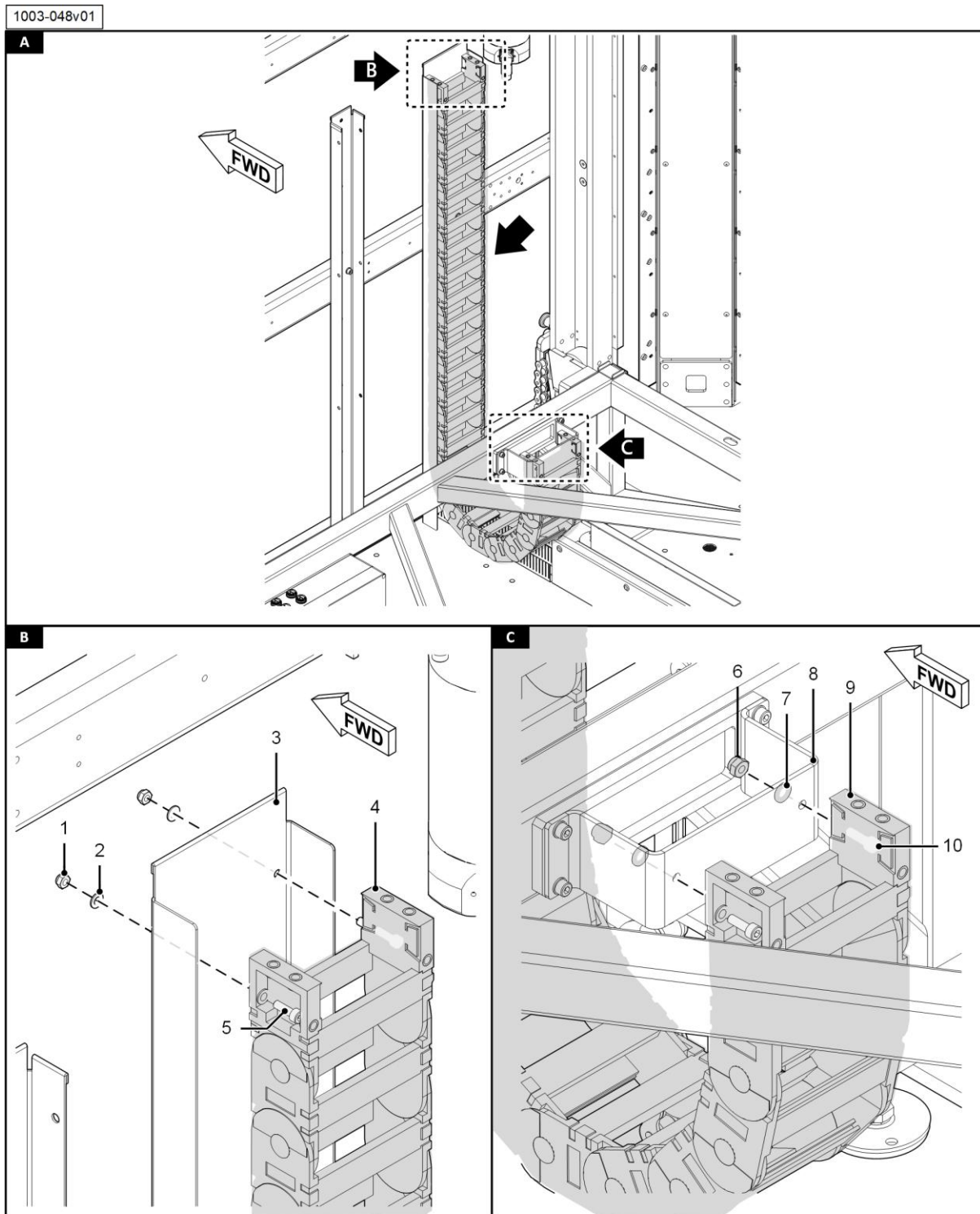


Figure 2-26: Igus Installation

2.4.13 Safe Sensor Cascade and Inductive Sensors

2.4.13.1 Part List

#	Figure No.	P/N	Description	Qty.	Type
1, 2	Figure 2-27	SL230172	Upper and Lower Horizontal Beams	-	Component
3	Figure 2-27	EC230057	Inductive Sensor, 4mm, PNP, NC, 10-30VDC	2	Component
4	Figure 2-27	Part of EC230057	M10 nut	-	Fastener
5	Figure 2-27	SL230172	Middle Horizontal Beam	-	Component
6	Figure 2-27	SL230141	Flexi loop holder	2	Fastener
7	Figure 2-27	FS230200	DIN7991 M4x10	4	
8	Figure 2-27	EC230080	Safe Sensor Cascade, Flexi Loop	2	Component
9	Figure 2-27	EC230160	Safe Sensor Cascade	1	Component

2.4.13.2 Installation Procedure

(Refer to Figure 2-27 below).

1. Install Top Inductive Sensor (3) on the Upper Horizontal Beam (1) as follows:
 - a. Insert the Inductive Sensor (3) into Top Horizontal Beam installation bore.
 - b. Tighten M10 nut (4) onto the Inductive Sensor (3) and thread the nut down until it reaches the Upper Horizontal Beam (1).
2. Repeat steps a to b for Bottom Inductive Sensor.
3. Position two Flexi Loop Holders (6) in its place on the Middle Horizontal Beam (5).
4. Using a 3 mm Allen-head bit, install four M4 Allen-head screws (7) which secure two Flexi Loop Holders (6) to the Middle Horizontal Beam (5).
5. Install Safe Sensor Cascade (9) into one Safe Sensor Cascade (8) MS connector.

Note

The Safe Sensor Cascade Flexi Loop (9) with the Safe Sensor Cascade (10) inserted into the left Flexi Loop Holder (6)

6. Insert two Safe Sensor Cascade Flexi Loop (8 and 9) into two Flexi Loop Holders (6).



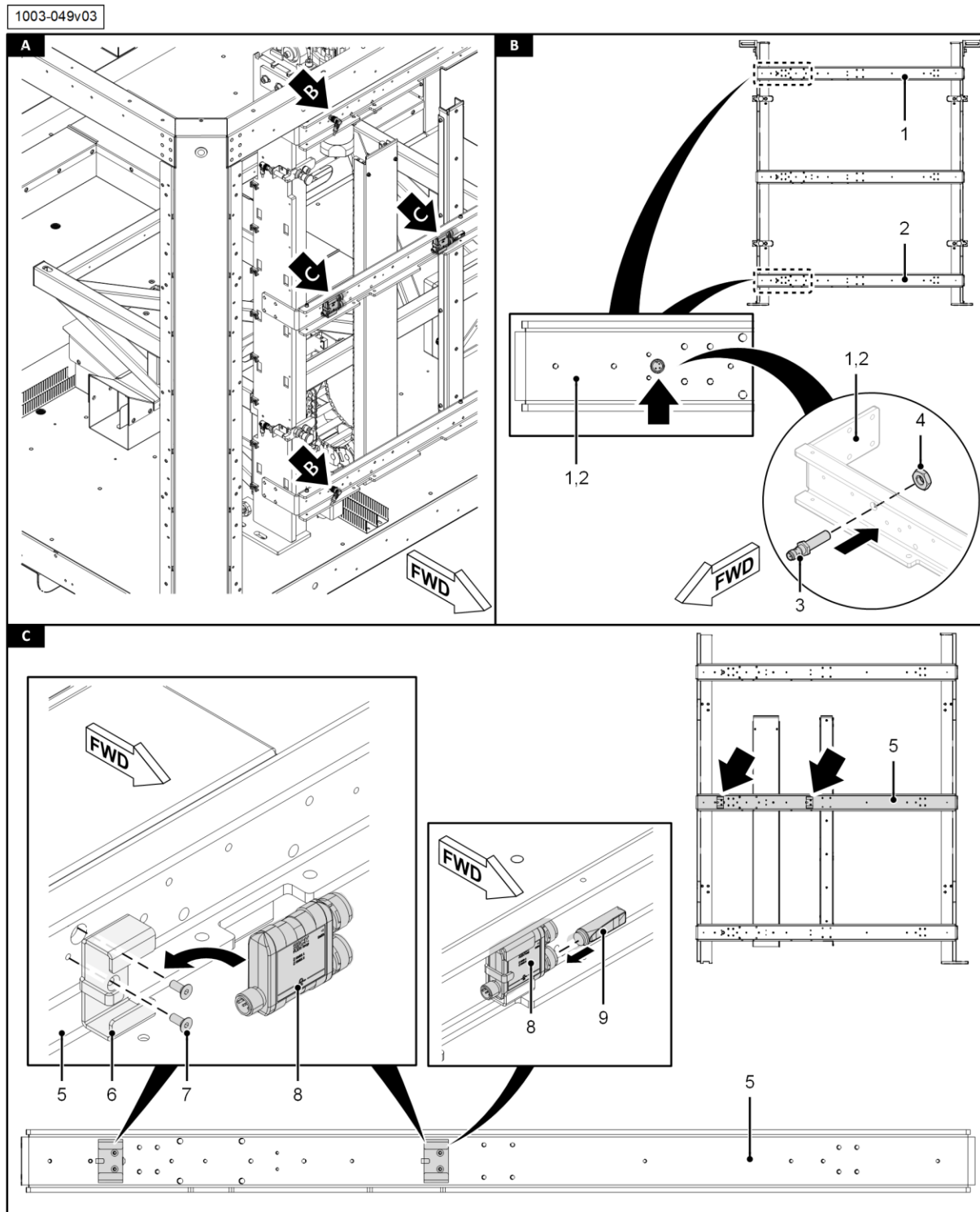


Figure 2-27: Safe Sensor Cascade and Inductive Sensors Installation

2.4.14 Lift Assembly Calibrations

(Refer to Figure 2-28 below).

1. Left and Right Roller Assembly Calibration:
 - a. Connect Motion Jig to the Drive Shaft Motor connector, for Motion Jig Connection refer to OPT1-D-063.
 - b. Lift the Lift Assembly 50 cm above the ground.
 - c. Measure the distance between the Roller Base (2) and the Side Beam (1).
 - d. Adjust the distance between Roller Base (2) and the Side Beam (1) to 3 mm as follows:
 - i. Using a 5 mm open-end wrench, release M6 nut (3).
 - ii. Using a 5 mm open-end wrench, adjust the distance between Roller Base (1) and the Side Beam (2) by turning the M6 Fastner Screw (4):
 - **Turn CW** to increase the distance.
 - **Turn CCW** to decrease the distance.
 - iii. Using a 5 mm open-end wrench, tighten M6 nut (3).
 - e. Repeat steps a to d for the other Roller Assembly (Left).

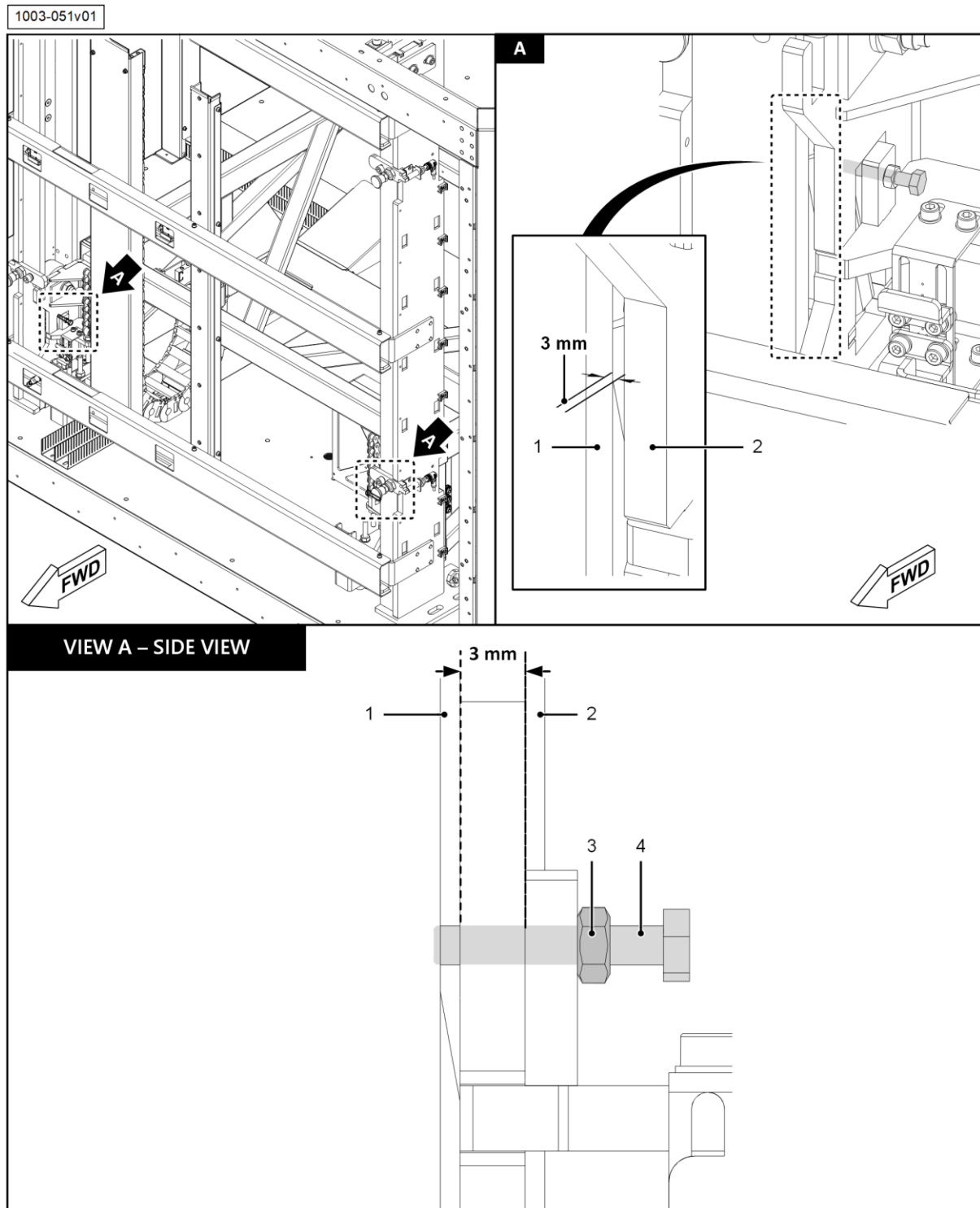


Figure 2-28: Lift Assembly Calibrations - Left and Right Roller Assembly Calibration

(Refer to Figure 2-29 below).

2. Inductive Sensors Calibration:

- a. Connect Motion Jig to the Drive Shaft Motor connector, for Motion Jig Connection refer to OPT1-D-063.
- b. Lower the Lift Assembly to the lower position.
- c. Using filler gauge, measure the distance between the Inductive Sensor tip (2) and the Flags Assembly (1).
- d. Adjust the distance between Inductive Sensor tip (2) and the Flags Assembly (1) to **1.25 mm $\pm$ 0.25 mm**:
 - i. Using a 17 mm open-end wrench, release two M10 nuts (3, 4), one from each side of the horizontal beam (5).
 - ii. Position Inductive Sensor tip (2) at **1.25 mm $\pm$ 0.25 mm** from the Flags Assembly (1).
 - iii. Using a 17 mm open-end wrench, tighten two M10 nuts (3, 4).
- e. Lift the Lift Assembly to the upper position.
- f. Repeat steps c and d for the upper Inductive Sensor.

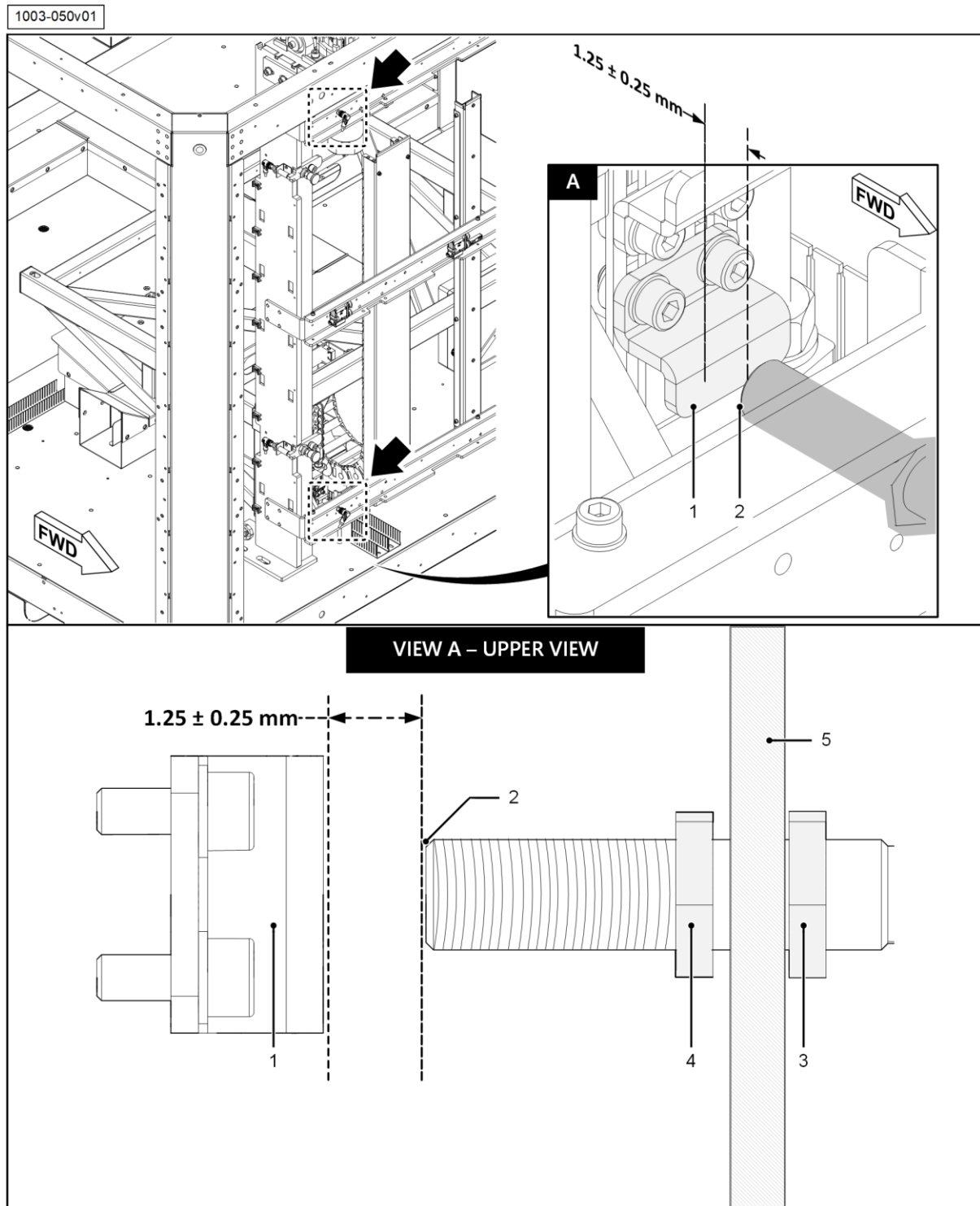


Figure 2-29: Lift Assembly Calibrations - Inductive Sensors